

My Working Experience on Music Research

Hsiao Wen Yi 蕭文逸

-- version 2024.09 --



AI Labs.tw
台灣人工智慧實驗室

Research Experience

 > **1K cites** (9 papers, 1 Journal)

 > **2.5K stars** (from 8 repositories) ID: wayne391

5 Yrs Working Exp, 7 Yrs in Music Research

'12

B.Sc.

Computer
Science

@Tsing Hua Uni.



'16

M.Sc.

Computer
Science

@Tsing Hua Uni.



'18

**Research
Assistant**

@Academia Sinica



'19

**Research
Engineer**

@Taiwan Allabs



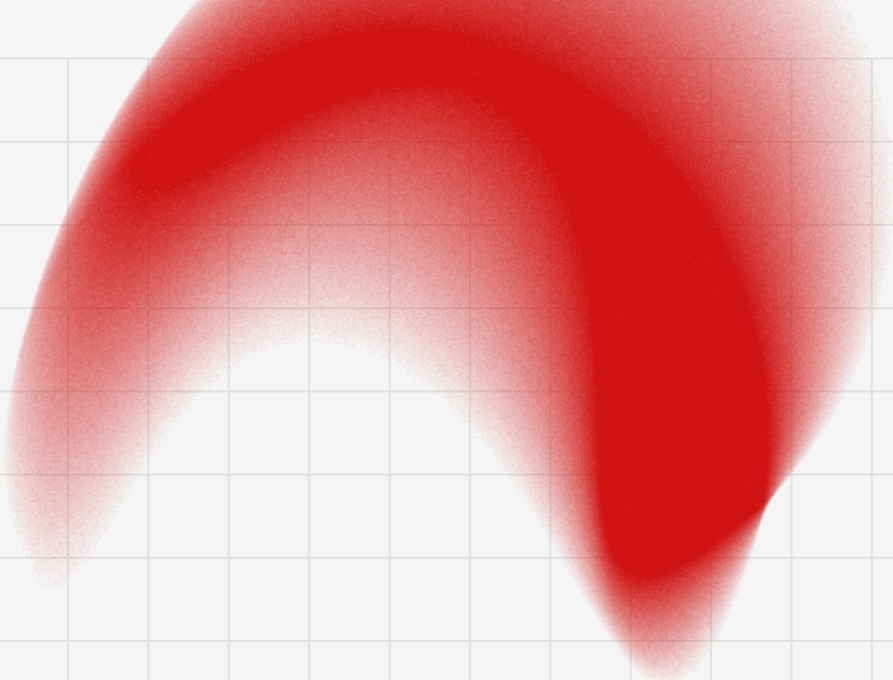
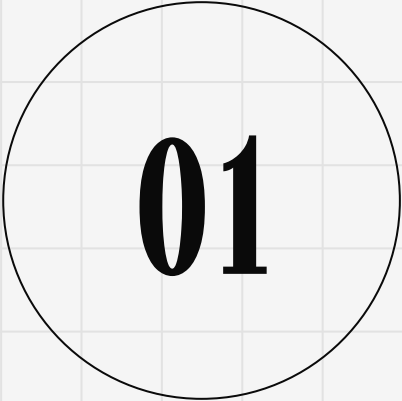
AI Labs.tw

台灣人工智慧實驗室

'24

**Research
Engineer
(Senior)**

**Master Thesis: Automatic Symbolic Music Generation Based on Convolutional GANs. (2018),
Adviser: Dr. Yi-Hsuan Yang.**



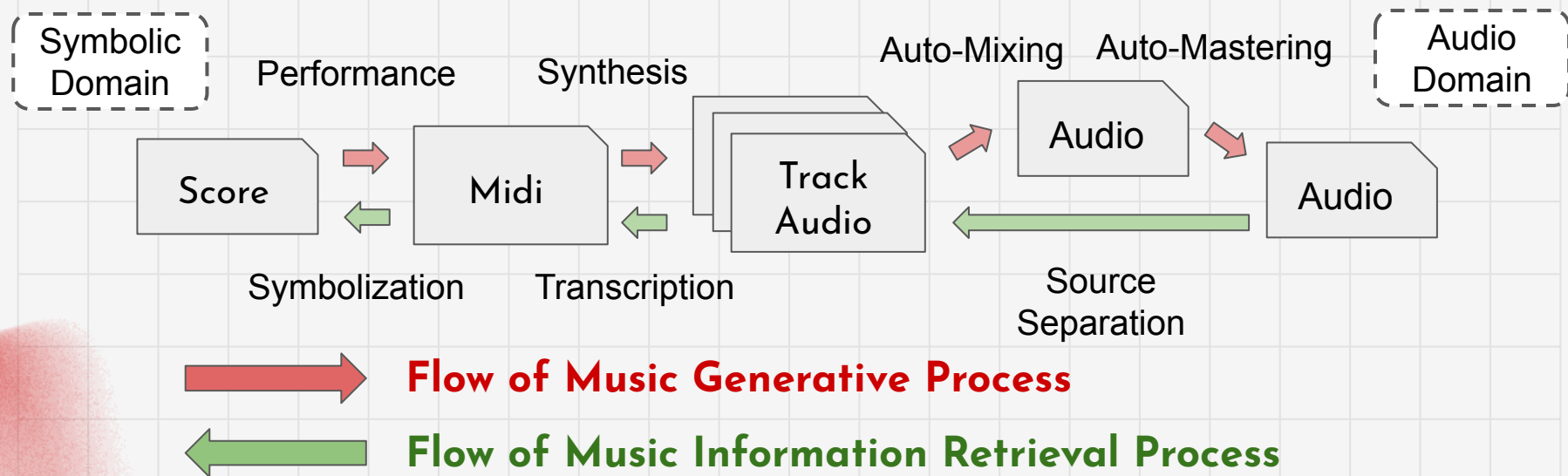
01

Work Overview

7-min Research Introduction

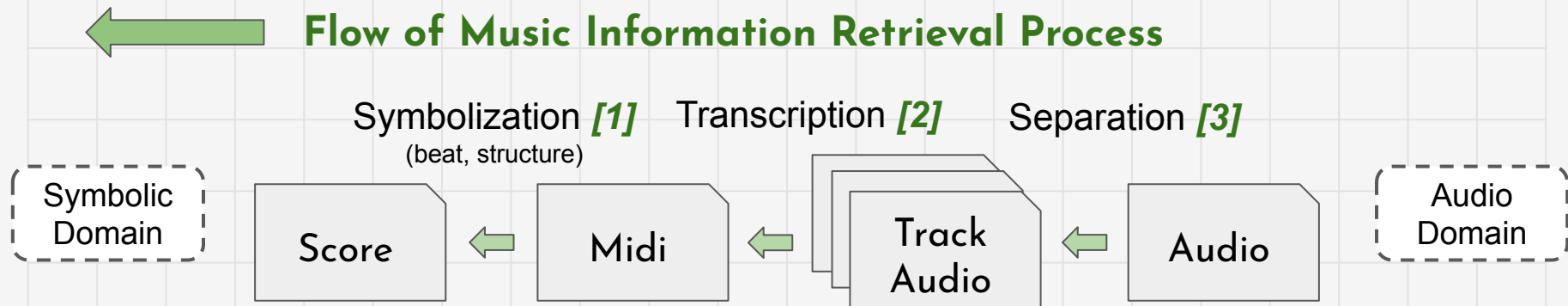
My Music Research Overview

- I have a **comprehensive experience** with the entire **Music Research pipeline**.
- I have strong knowledge of modern **music production industry**.
- I have **cross-domain** (score, text and audio) modeling experience.
- In what following, I will **prove my skill** by **publications** and **github repos**



My Music Research Overview

Flow of Music Information Retrieval Process



(ISMIR'22, 2nd author) **Transcription of Polyphonic Electric Guitar Music** [2]

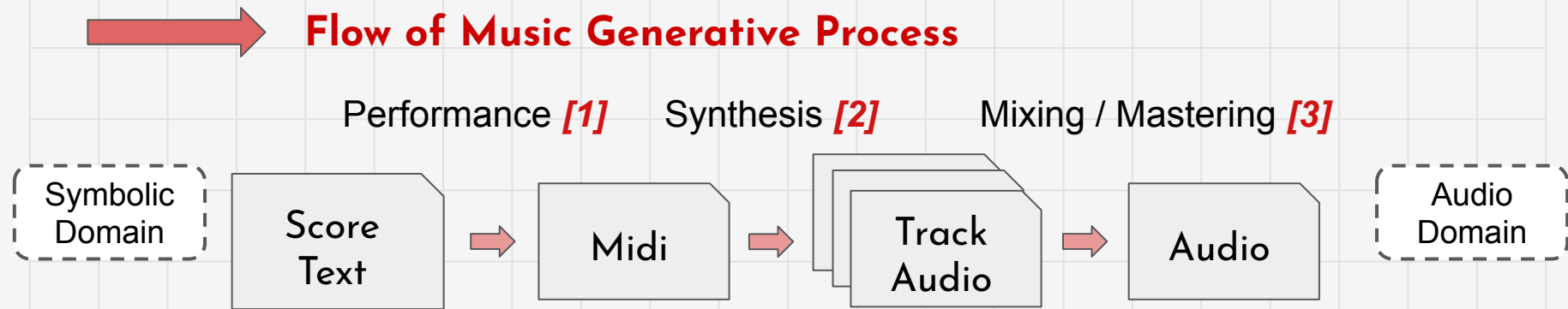
(Eusipco'21, 3rd author) **Beat and Downbeat Tracking Enhanced with Source Separation** [1]

(MMSP'20, 2nd author) **Blind Violin/Piano Source Separation with Mixing-specific Data Augmentation** [3]

- **MidiToolkit** [1] (227 stars) - Popular and Fundamental Tool for MIDI Processing
 - Conversion between absolute & symbolic timing
- **SF Segmenter** [1] (52 stars) - Structure Analysis with Structural Feature

My Music Research Overview

Flow of Music Generative Process



(ISMIR'24, 2nd author) **MusiConGen** - Text-to-Music Audio Generation with Chord and BPM Control **[1,2,3]**

(DAFX'24, 2nd author) **Hyper RNN for AFx Modeling** **Oral** **[3]**

(ISMIR'22, co-1st author) **DDSP-based Singing Vocoders** - Differentiable DSP Singing Vocoder **[2]**

(AAAI'21, 1st author) **Compound word transformer** - Symbolic Music Generation for piano **[1,2]**

(ISMIR'21, 3rd author) **Guitar Tabs by Transformers and Groove Modeling** **[1,2]**

(AAAI'18, co-1st author) **Musegan** - Symbolic Music Generation **Oral** **[1]**

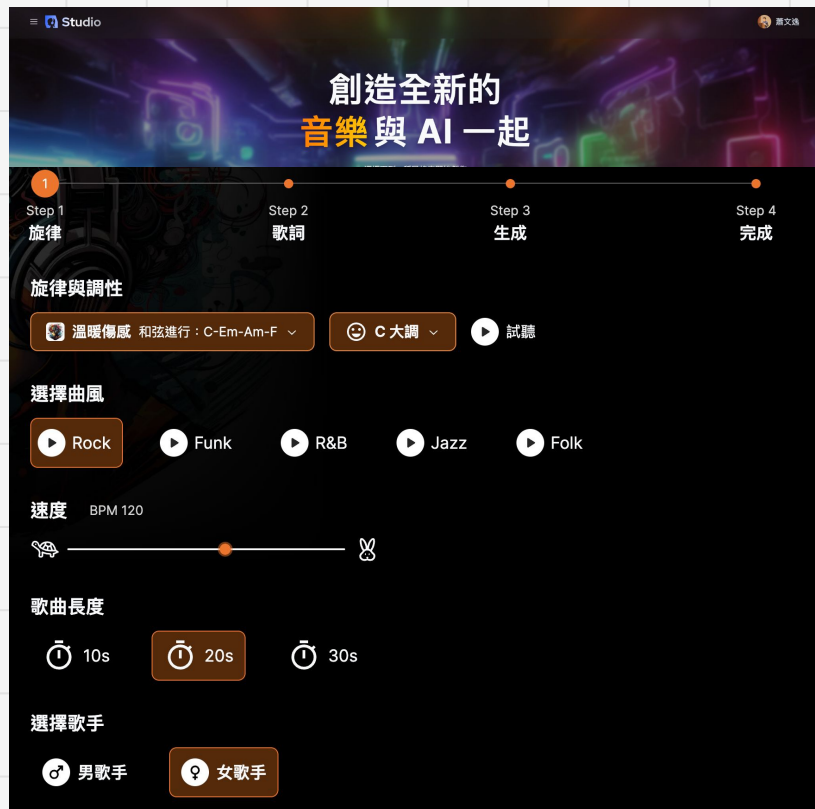
Production

教我如何做你的愛人 - 陳珊妮 AI 模型

Collaboration with a Popular Chinese Singer



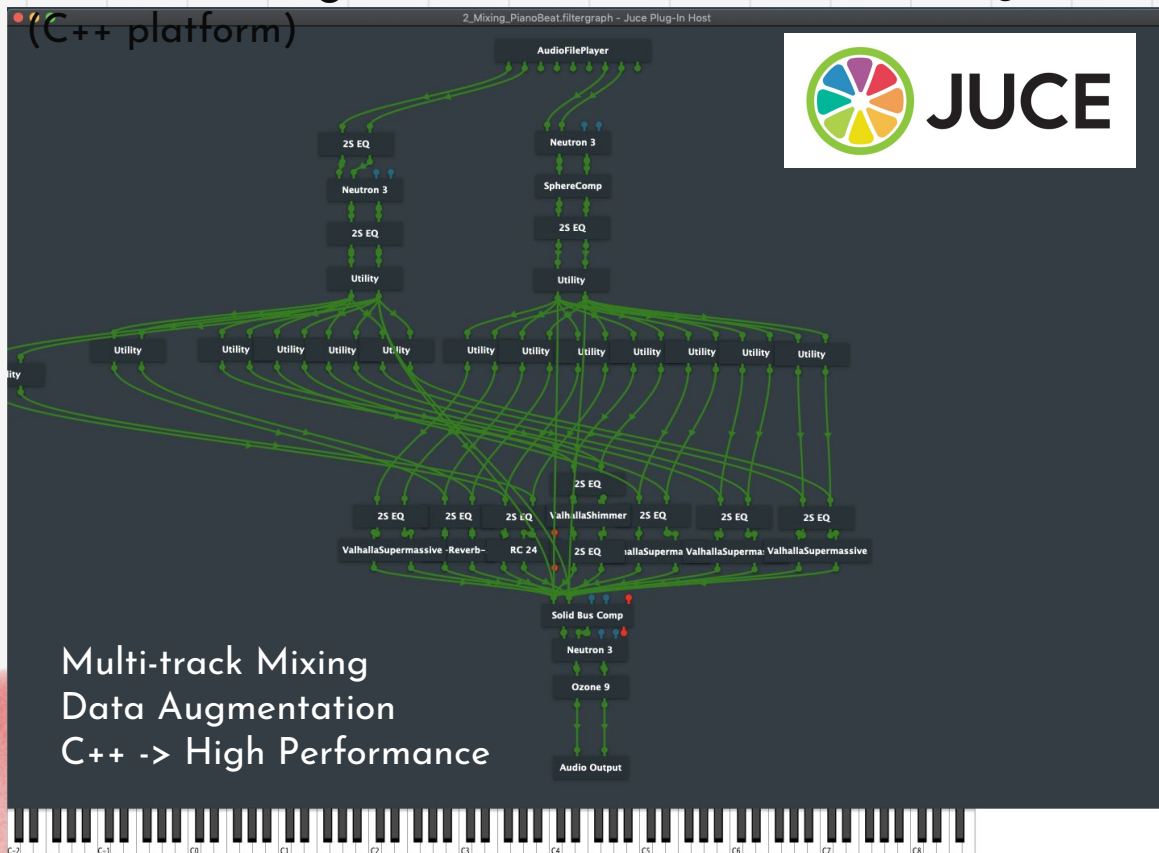
AI Singer + AI Song Maker



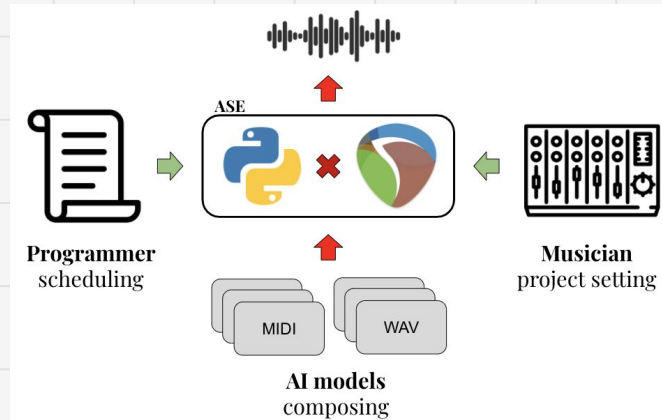
Yating Music - Song Creation Platform

Inhouse Mixing Backend, Based on JUCE C++ Plugin Host

Inhouse Mixing Backend, Based on JUCE C++ Plugin Host



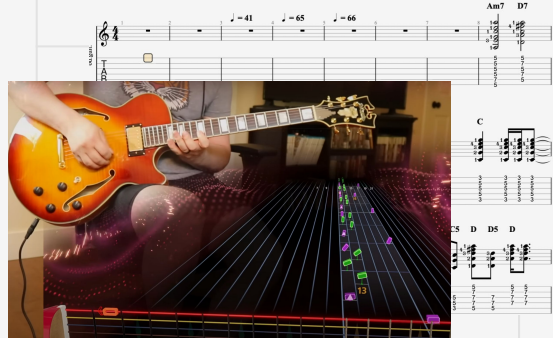
- OpenSource:
 - **ReaRender** (94 stars)



Dataset Building

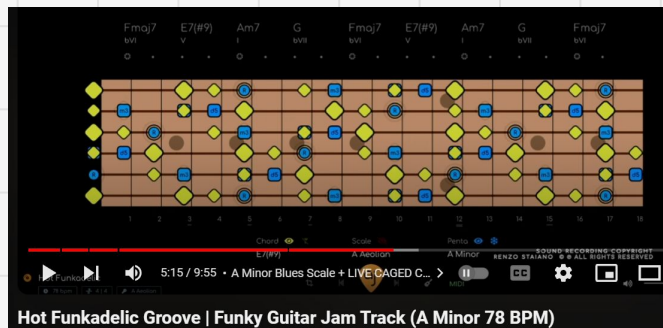
Highlights of My Inhouse Collection:

1. Data from Guitar Gaming Community



- Aligned audio and tab
- Finger position
- Chord label
- Over 1K songs
- Multi-track guitar
- Tab Generation
- Transcription

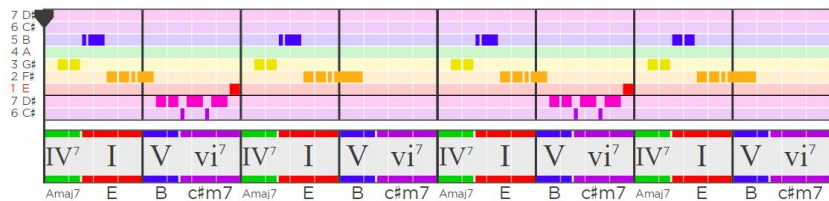
3. Backing Tracks



Skillset:

- Web Crawling, Data Cleaning
- Musicology

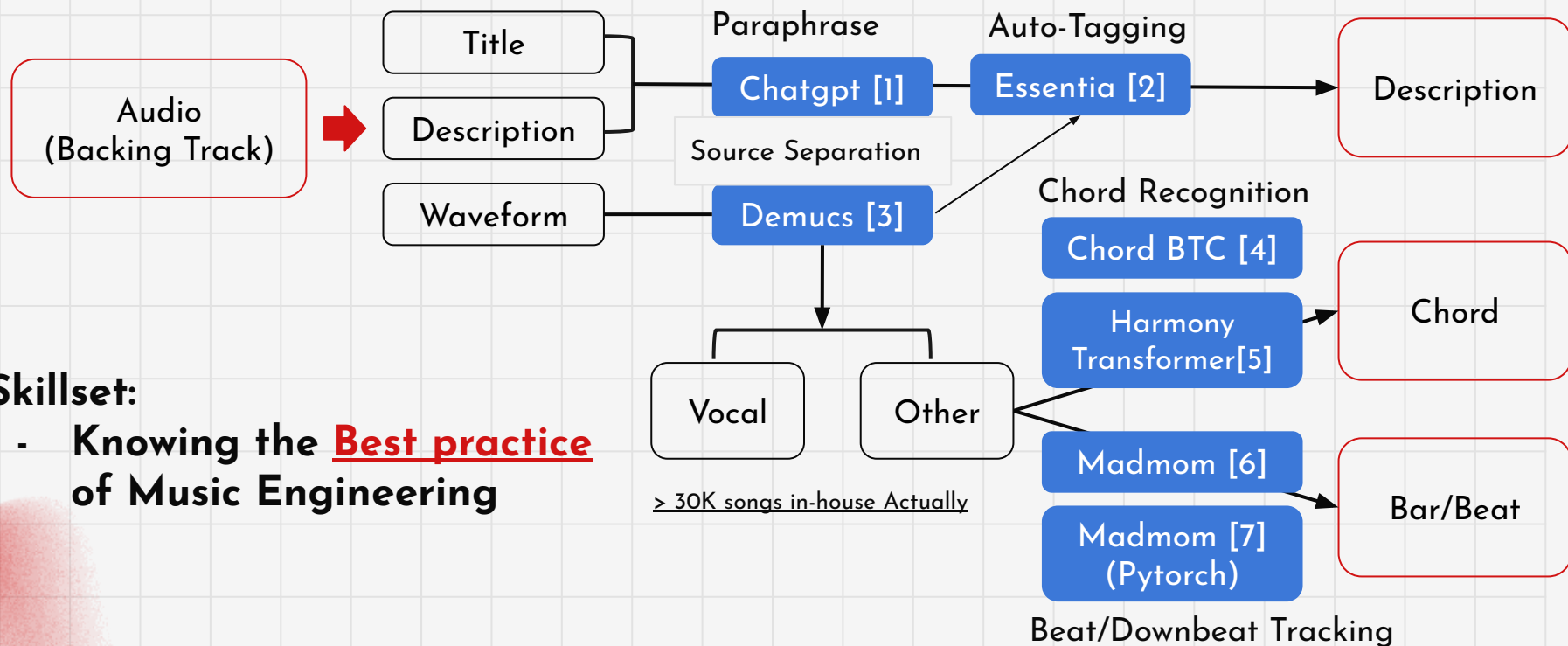
2. Lead Sheet from theorytab (108 stars)



- Over 30k songs
- Our backbone dataset of text2music model
- Description
- Key
- BPM
- Chord Progression
- High Quality after Curation (TODO)
-> Excellent Resources for any task!

Dataset Building

- Pipeline from my work - MusicConGen (ISMIR'24)



Skillset:

- Knowing the Best practice of Music Engineering

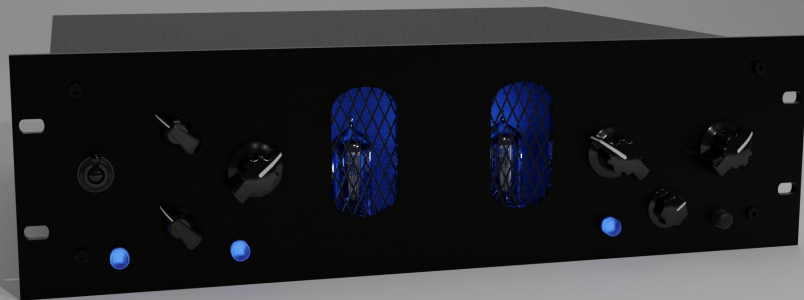
Side Projects

Audio Effect Emulation with AI & Make EQ/Distortion Plugin with JUCE

- TorchLite Demo (5 stars)
- Similar Product:
 - Neural DSP, Positive Grid, ...



Mixing Gear (vacuum tube) Emulation



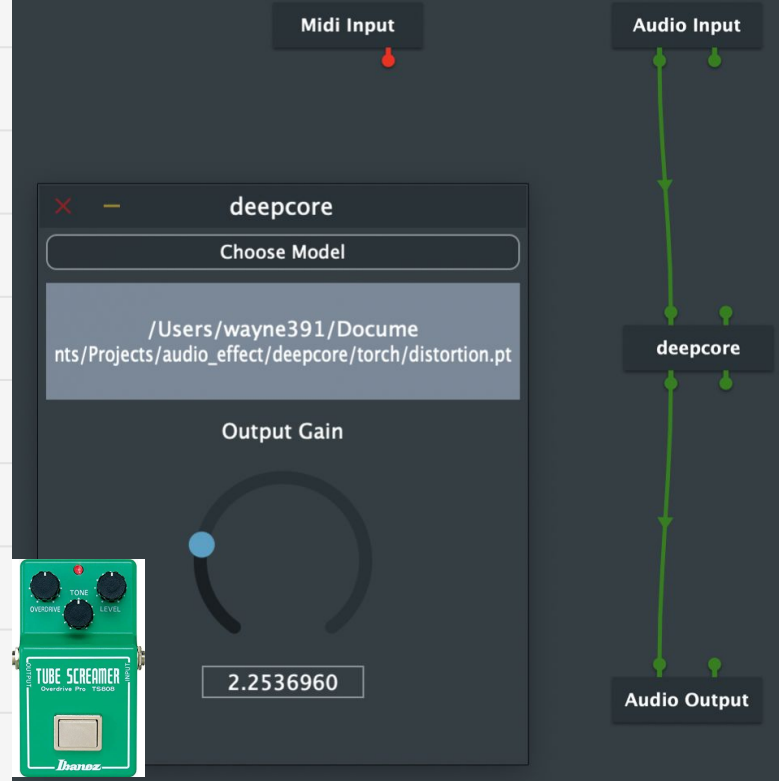
My 3D Modeling Artwork :D

(DAFX'24, 2nd author) Hyper RNN for AFx Modeling

Skillset:

- Train DSP-inspired NN Models
- Deploy with C++ (Libtorch + Eigen)

TS-808 Pedal Real-time Emulation



Visibility

Build Open-source Ecosystem of Our Company



Yating Music, Taiwan AI Labs

A research team working on Music AI technology at the

86 followers

Taipei, Taiwan

<https://ailabs.tw>

Popular repositories

remi

Public

"Pop Music Transformer: Beat-based Modeling and Generation of Expressive Pop Piano Compositions", ACM Multimedia 2020

Python 540 84

ddsp-singing-vocoders

Public

Official implementation of SawSing (ISMIR'22)

Python 249 35

MuseMorphose

Public

PyTorch implementation of MuseMorphose (published at IEEE/ACM TASLP), a Transformer-based model for music style transfer.

Python 170 32

compound-word-transformer

Public

Official implementation of compound word transformer (AAAI'21)

Python 265 43

miditoolkit

Public

<https://pypi.org/project/miditoolkit/>

Python 227 35

ReaRender

Public

A python toolkit for automatic audio/MIDI rendering using REAPER

Python 94 16



Wen-Yi Hsiao

ailabs

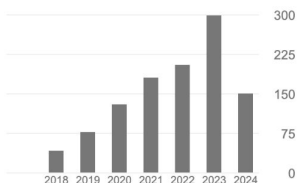
在 ailabs.tw 的電子郵件地址已通過驗證

[Machine Learning](#)

追蹤

引用次數

	全部	自 2019 年
引文	1097	1048
H 指數	10	10
i10 指數	10	10



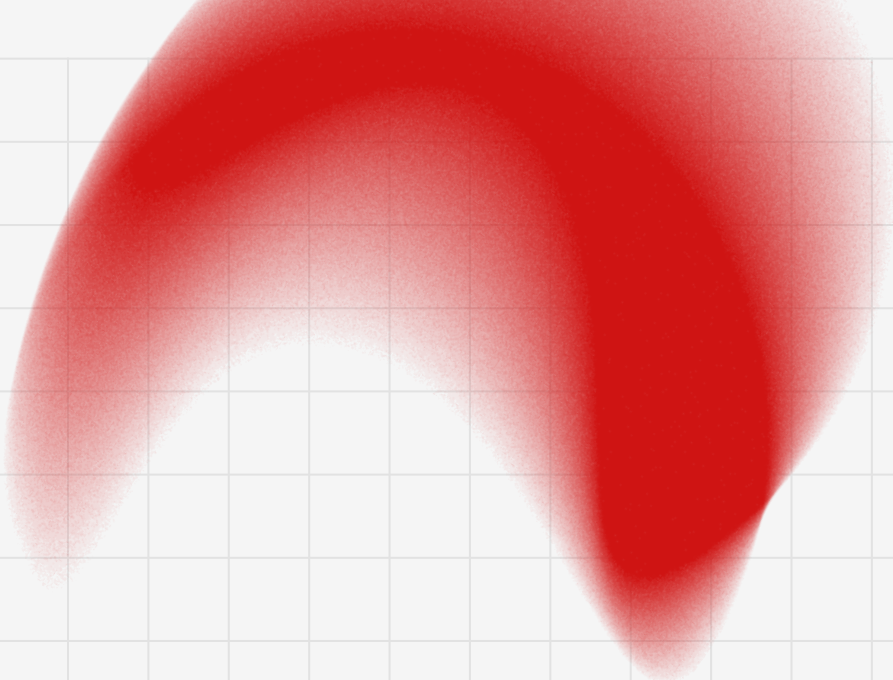
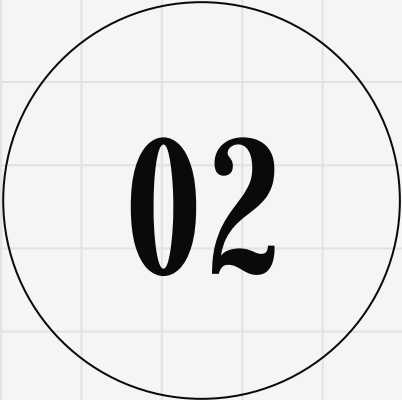
當AI遇見數位音樂－創作工具的介紹與應用

27 9 月, 2023

當AI
遇見數位音樂：
創作工具
的介紹與應用

2 @ 112A — NYCU
3 跨域系列講座

Product Promotion -
Campus Workshop @NYCU



02

Audio To Symbolic Domain

Music Information Retrieval (MIR)

Audio to Symbolic Domain

What is the Symbolic Domain in Music?

Human understand music with notations and the conceptualized **informations**:

- BPM
- Meter
- Lead Sheet
 - Key
 - Chord
 - Melody
- Arrangement
- Structure
- MIDI
- Sheet Music
 - Staff and Tablature
- Genre
- Description (Autotagging)

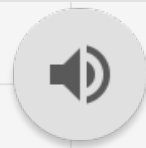
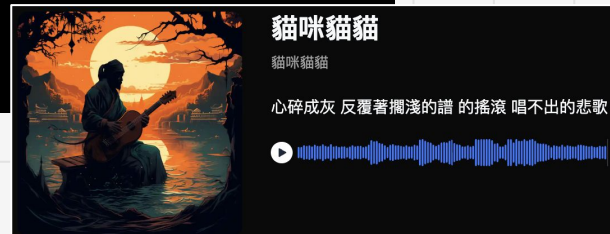
What are the Models to extract theses infos?

Why?

1. For GenAI: Understand then can control
2. Recommendation System
3. Human readable format (Transcription)

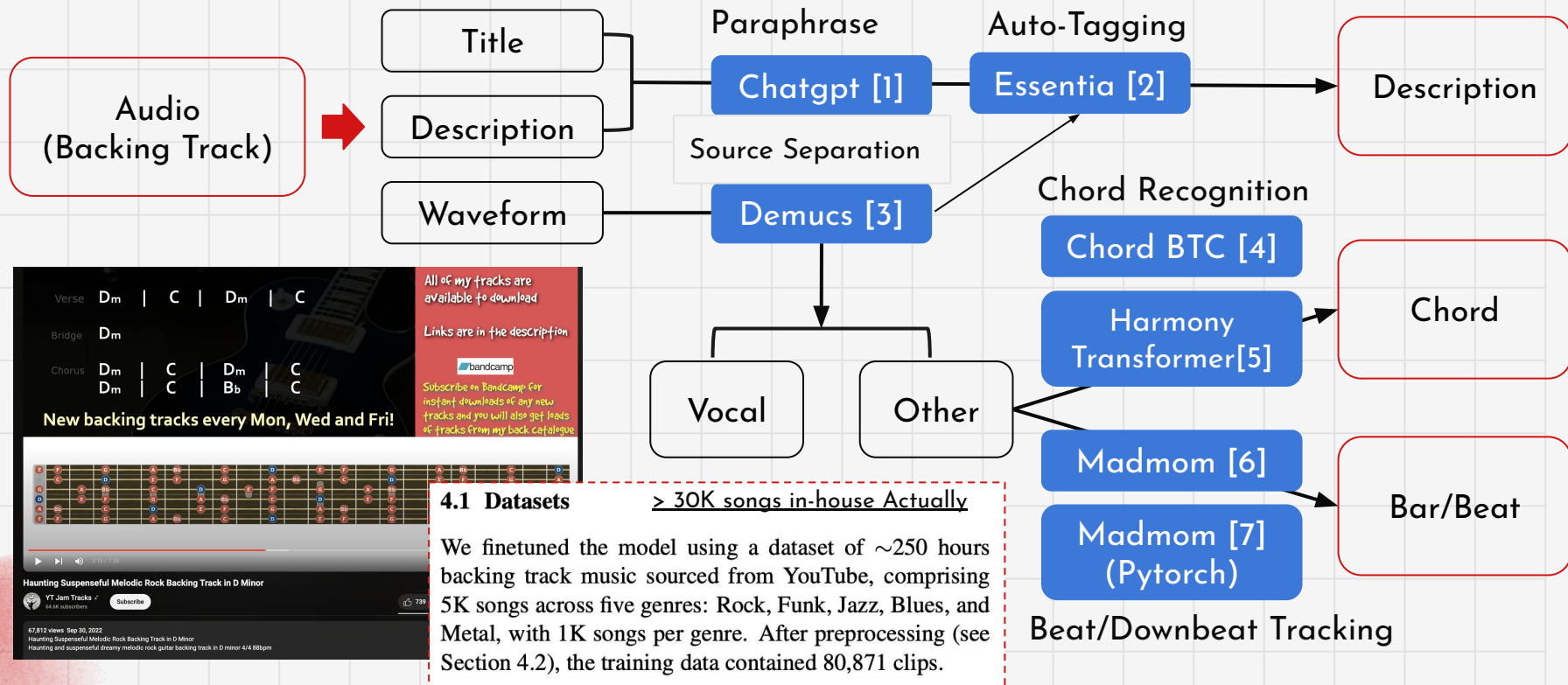


MusicConGen
(ISMIR'24)



Audio to Symbolic Domain - Example I

- MusicConGen (ISMIR'24) - Data preprocessing Pipeline**



Therefore, we have the pair (Non-Vocal Audio, Text, Chord, Beat/Downbeat) as the training data

Audio to Symbolic Domain - Example I

References:

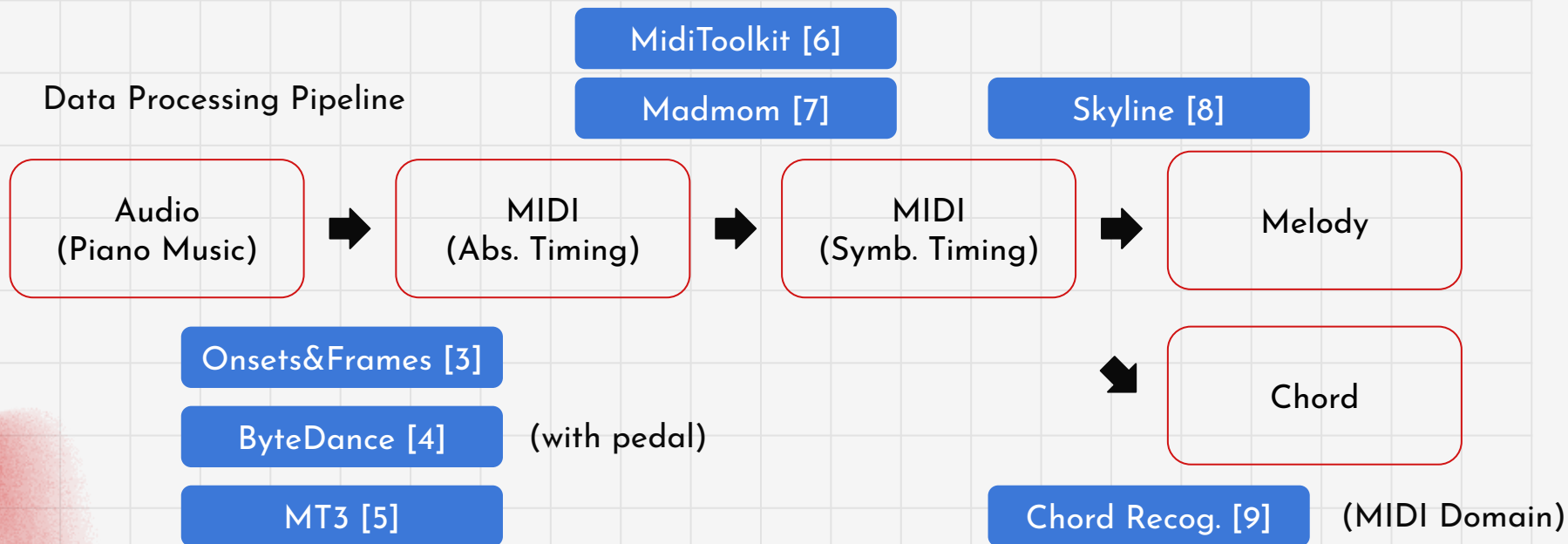
- [1] [ChatGPT API](#)
- [2] [MTG/essentia](#)
- [3] [facebookresearch/demucs](#)
- [4] [jayg996/BTC-ISMIR19](#)
- [5] [Tsung-Ping/Harmony-Transformer-v2](#)
- [6] [CPJKU/madmom](#)
- [7] [ben-hayes/beat-tracking-tcn](#)

Sample File:

```
{
  "key": "G",
  "artist": "",
  "sample_rate": 48000,
  "file_extension": "wav",
  "description": "",
  "keywords": "",
  "duration": 30.0,
  "bpm": 112,
  "genre": "Rock, Gothic Metal, Death Metal, Doom
    Metal, Goth Rock, Melodic Death Metal, Progressive
    Metal, Heavy Metal",
  "title": "",
  "name": "",
  "instrument": [
    "drums",
    "electricguitar",
    "bass",
    "guitar",
    "synthesizer",
    "voice",
    "keyboard"
  ],
  "moods": "epic, dark, melodic, heavy, energetic,
    sad",
  "path": "./_4mH2HwVF-0/13/no_vocal.wav"
}
```

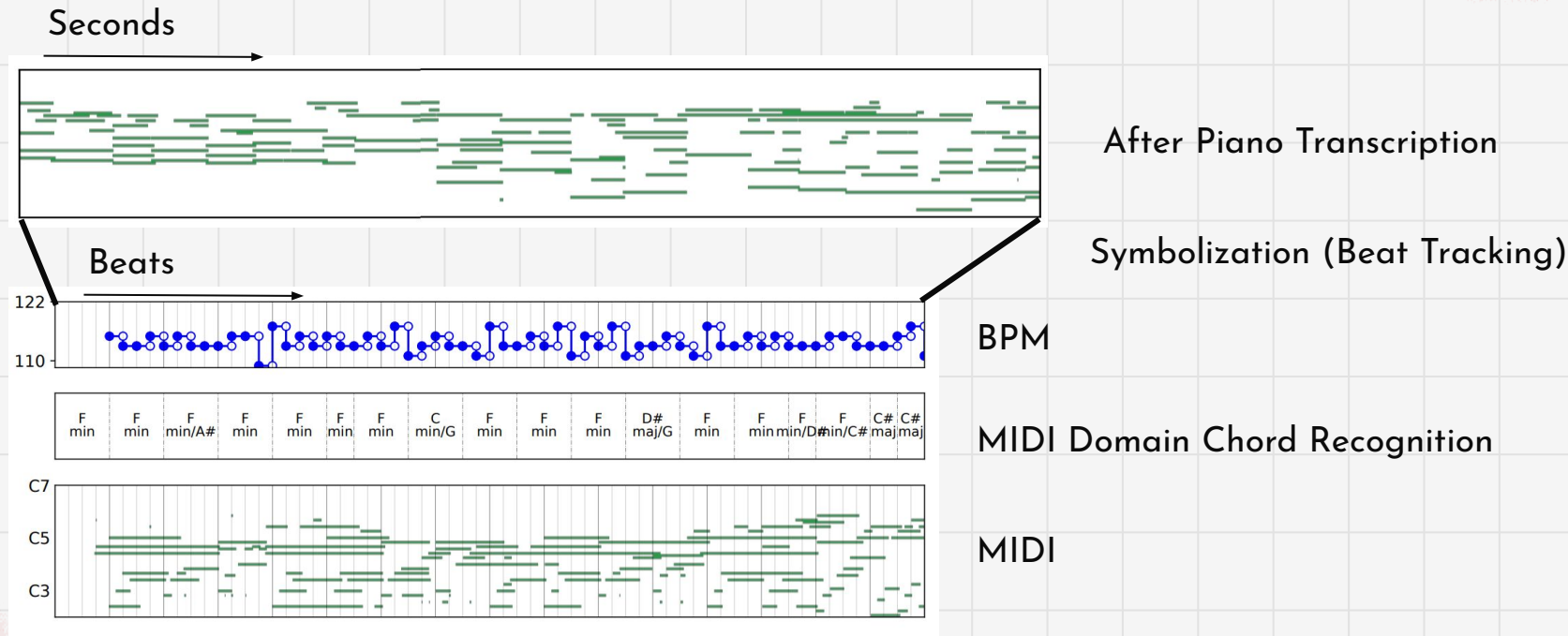
Audio to Symbolic Domain - Example II

1. **Goal:** {Piano MIDI, Lead Sheet} x {Transcription, Generation}
 - a. Compound Word Transformer (AAAI'21) [1]
 - b. REMI (ACMM MM'20) [2]



Audio to Symbolic Domain - Example II

Explain: Timing Symbolization with Miditoolkit [6] and Madmom [7]



MIDI Chord Recognition Toolkit: Choder (91 starts)
developed by me and our former intern

Audio to Symbolic Domain - Example II

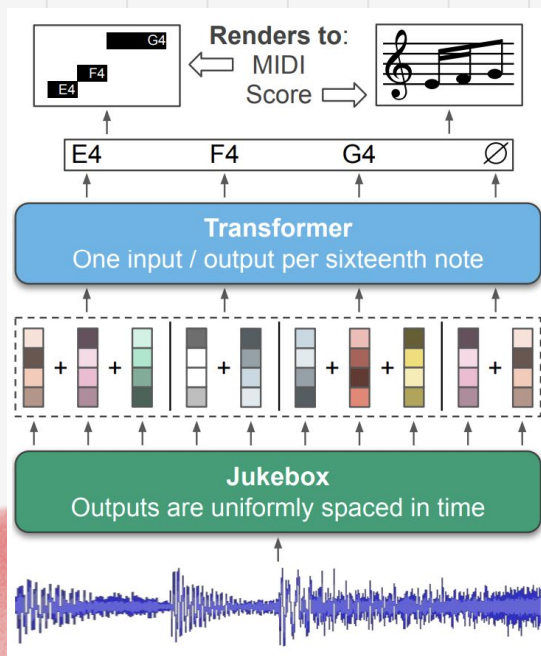
References:

- [1] [YatingMusic/compound-word-transformer](#)
- [2] [YatingMusic/remi](#)
- [3] [jongwook/onsets-and-frames](#)
- [4] [bytedance/piano_transcription](#)
- [5] [magenta/mt3](#)
- [6] [YatingMusic/miditoolkit](#)
- [7] [CPJKU/madmom](#)
- [8] [MIDI-BERT/tree/CP/melody_extraction/skyline](#)
- [9] [joshuachang2311/chorder](#)

Audio to Symbolic Domain - Example III

Goal: Lead Sheet Generation

- SheetSage on 20K in-house curated pop song



What a Wonderful World

George David Weiss and Bob Thiele

J = 69

1 F A- B♭7 A-7 G-7 F A7 D- *2*

I see trees of green, red-ros-es too, I see them bloom for me and you, and I

3 D♭ C[#] C7 F F+ B♭Δ7 C7

think to my-self WHAT A WON- DER- FUL WORLD I see

1 F A- B♭ A-7 G-7 F *2*

skies of blue and clouds of white, the bright bless-ed day, the

3 A7 D- D♭ C[#] C7 F B♭ F

dark sac-red night, and I think to my-self WHAT A WONDERFUL WORLD The

1 C7 F C7 F *2*

col-ors of their ain-bow, so pret-ty in the sky are al-so on their fac-es of peer-ple-gin'-by, I see

3 D- C/E D-7 F C/G D-7 F#m7 G-7 F#m7 C7

friends shak'-in' hands, say-in', 'How do you do!' They're ree-ah say-in' 'I love you, 'I hear

1 F A- B♭ A-7 G-7 F A7 D- *2*

ba-bies cry I watch them grow, learn-much more than I'll ev-er know, and I

3 D♭ C7 C7 F A-7 D7 G-7

think to my-self WHAT A WON- DER- FUL WORLD Yes, I think to my self

1 C7 F B♭ F

WHAT A WON- DER- FUL WORLD

1927

Chord is the key to our all service

- **Chord** to Vocal Melody
- **Chord** + Melody to Piano MIDI
- **Chord** + Text to Music

“Chord” can make individually generated tracks sound harmonic

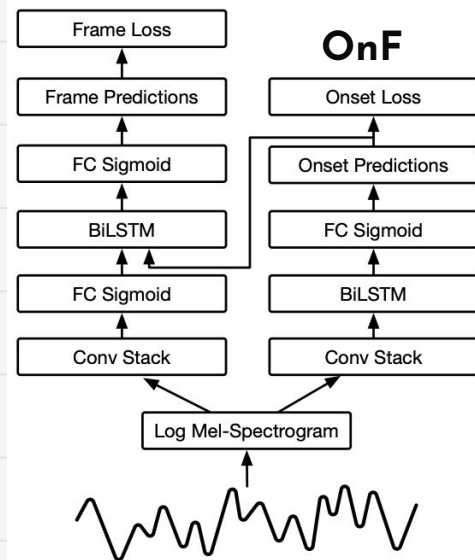
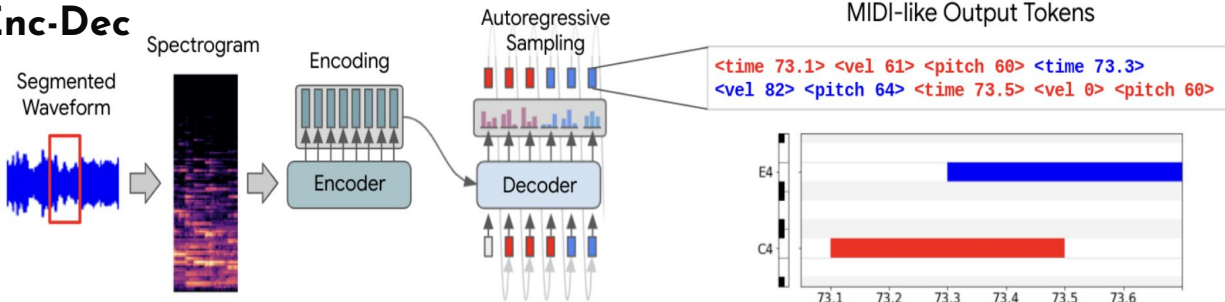
- Sheetsage Problem
 - Extremely Slow
 - Jukebox Pretrained Feats

Audio to Symbolic Domain - Example IV

Transcription - Transcribe Audio into MIDI

1. Onset and Frames (Onf)
(by Curtis Hawthorne, ISMIR'17)
2. Sequence-to-Sequence Piano Transcription with Transformers
(by Curtis Hawthorne, ISMIR'21)
3. MT3: Multi-Task Multitrack Music Transcription
(by Josh Gardner, ICLR'22)

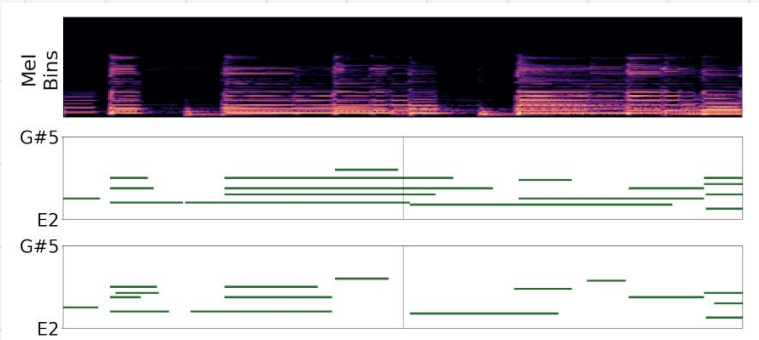
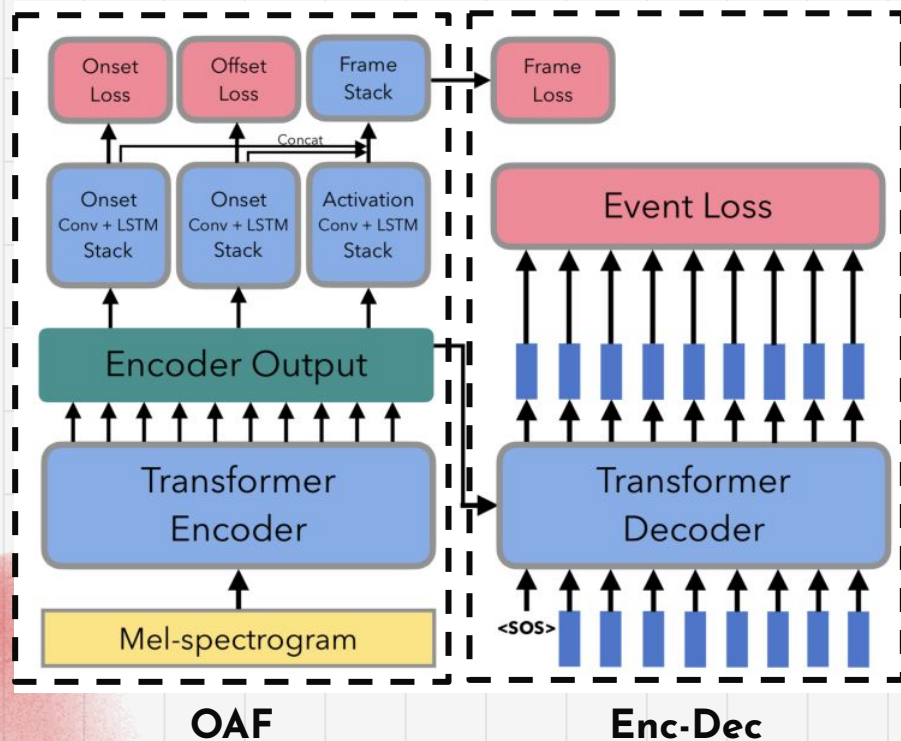
Enc-Dec



Audio to Symbolic Domain - Example IV

Inspired by (1), (2) and (3)

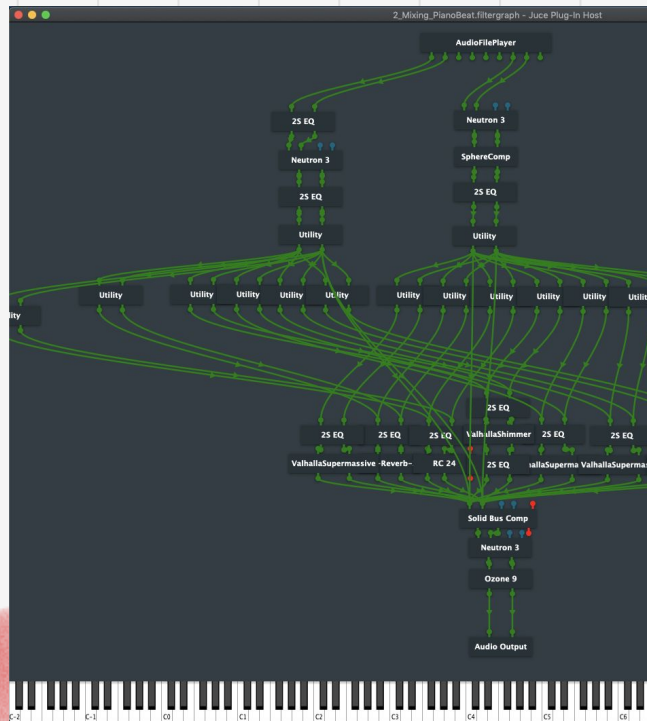
We proposed a Novel Guitar Transcription Model (ICASSP'22)



Model	(Encoder output)	
	Onset F1	Frame F1
Onset and Frame (OAF) [14]	0.591	0.583
CE-only Transformer [17]	0.543	0.523
	0.554	0.524
	0.568	0.537
Proposed multi-loss Transformer	0.598	0.579
	0.604	0.573
	0.613	0.582

Audio to Symbolic Domain - Example IV

... and a new Guitar Dataset - EGDB



DI (Direct Input)
Clean Signal



Guitar Rig Plugin

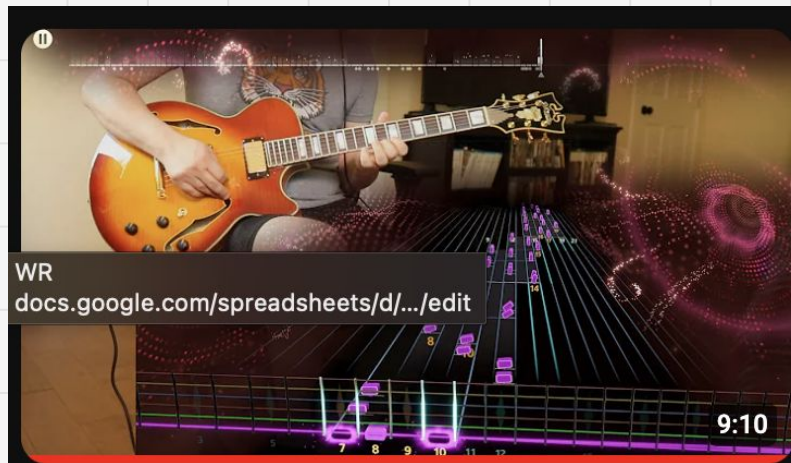
Colored Singal

- The DI (input signal) is recorded by musician
 - Given a Tab
 - Sight Reading Performance
 - w/ a special pickup
 - Post-processing
 - Human Curation
 - Rendered by JUCE
 - w/ different tones
- We have (tab, DI, color) pairs
 - Audio of individual string



Audio to Symbolic Domain - Example IV

Current Plan on Guitar, a Larger Dataset



"Master of Puppets" Metallica - Lead Rocksmith+

觀看次數：54萬次 · 1 年前



Riff Repeater

Available Worldwide! not only have we lived to see moar muse in rb, but there's officially Metallica in **Rocksmith**. Albeit in a...

4K

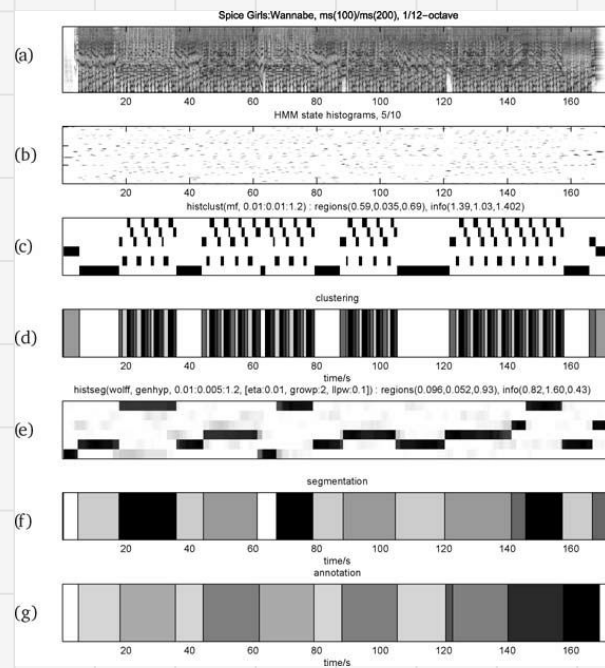
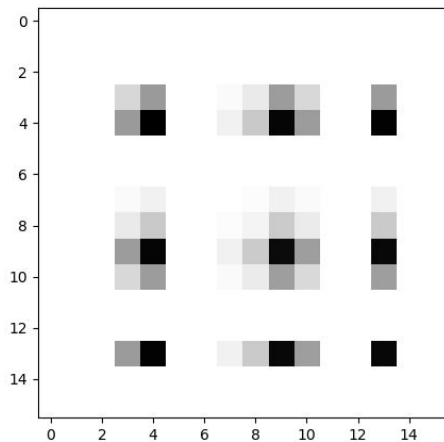
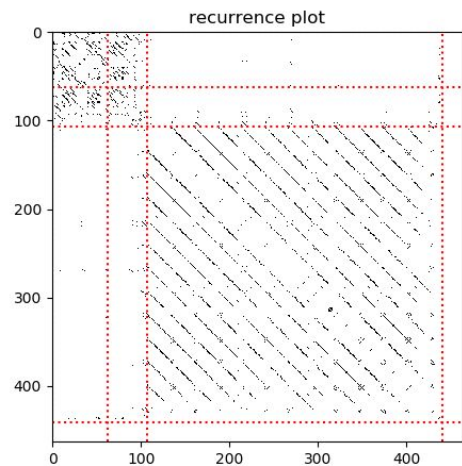
Our Vision:

- Aligned audio and tab
- Tab, not only MIDI (Position & Fingering)
- Chord label
- Over 1K songs
- Multi-track guitar
- Transcription

Audio to Symbolic Domain - Example V

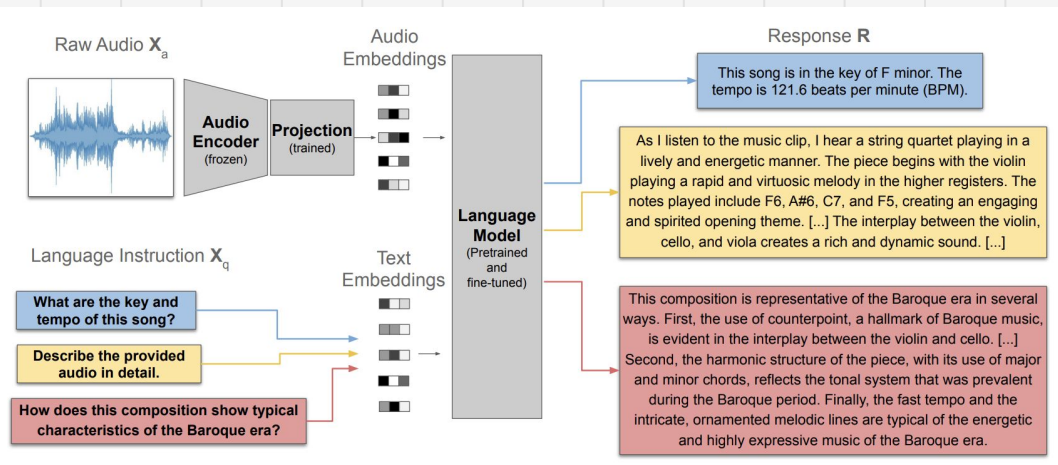
Structure = Boundary + Section Labeling

- MSAF Toolkit, by Oriol Nieto, ISMIR'16
- Unsupervised Music Structure Annotation w/ Structure Features (SF)
 - Joan Serrà, AAAI'16, IEEE MM'17
- SF Segmenter (by me, 52 stars), works on **MIDI & Audio**



Audio to Symbolic Domain - Example VI

- LLark (from spotify)



- MERT

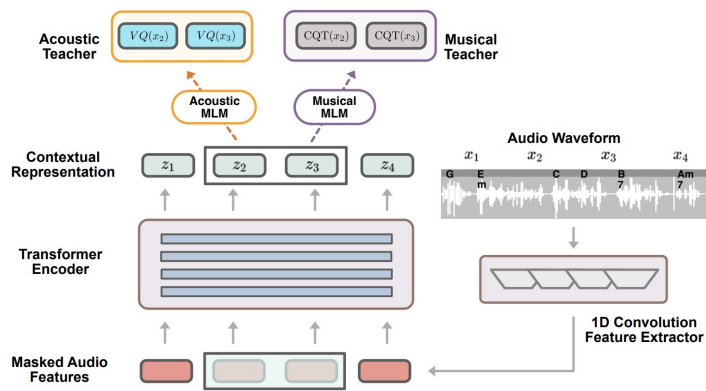


Figure 1: Illustration of the MERT Pre-training Framework.

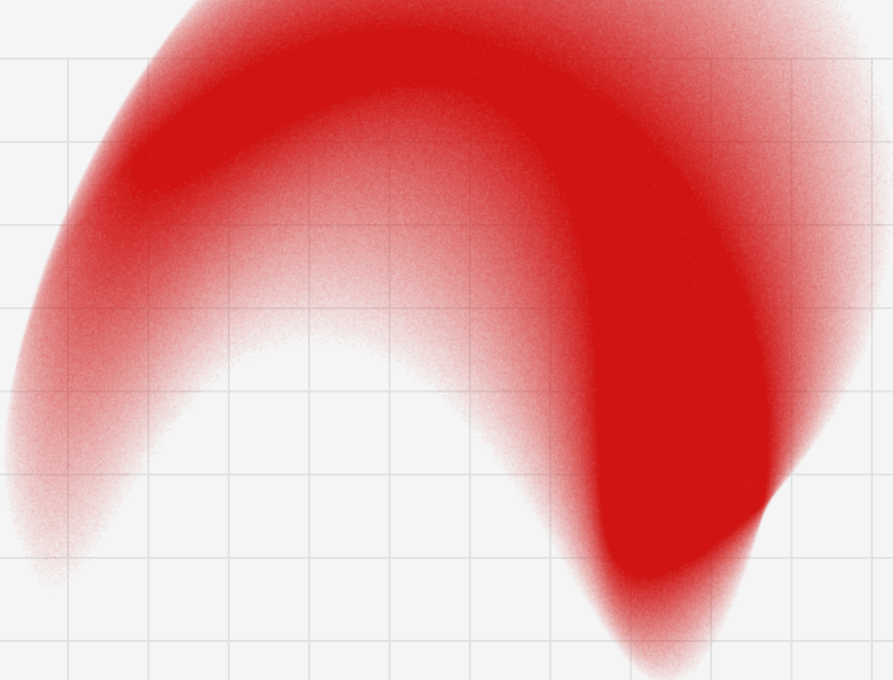
- Not enough resources (especially GPU) in my current company : (
- But... Rethinking the necessity?
- If there are enough resources, I can do scaling with my expertise :)

Audio to Symbolic Domain

Quick Review of My MIR Tech Stack

- BPM Madmom
- Meter
- Lead Sheet SheetSage Crepe
 - Key Chord BTC
 - Chord
 - Melody Skyline
- Arrangement Demucs
- Structure MSAF
- MIDI
- Sheet Music Transcription Onf/MT3/...
 - Staff and Tablature
- Genre ChatGPT Essentia
- Description

Baseline of All - **Essentia from MTG**, an old but **fast** universal Auto-Tagging model



03

Symbolic To Audio Domain

Generative AI Music

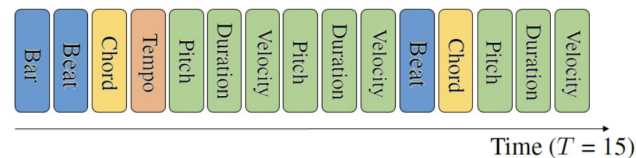
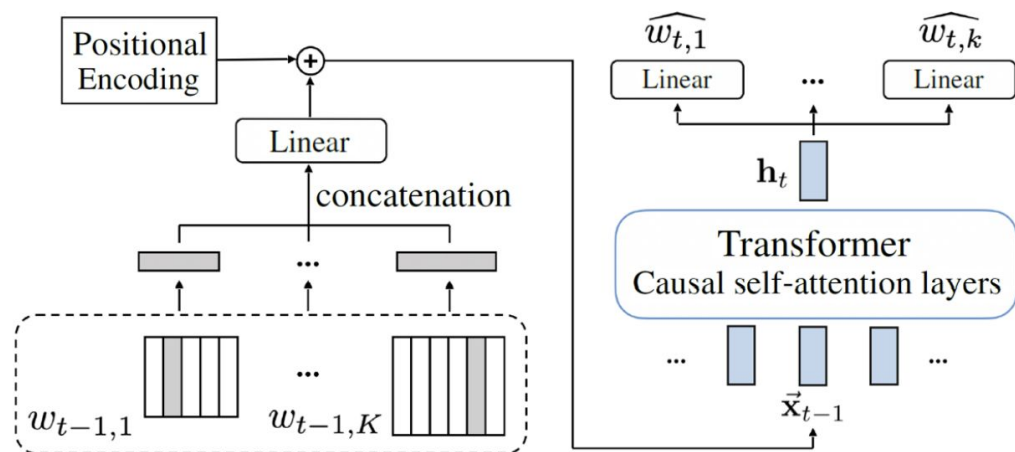
Symbolic to Audio to Domain - Example I

Goal: Generate Piano MIDI (Symbolic Domain Generation)

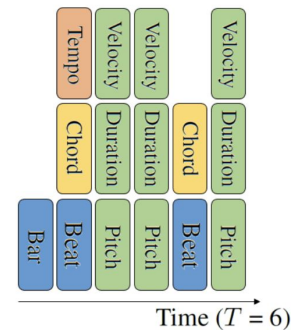
- Compound Word Transformer** (AAAI'21) [1] | DEMO
- REMI** (ACM MM'20) [2]

MIDI Note = Pitch + Duration + Velocity

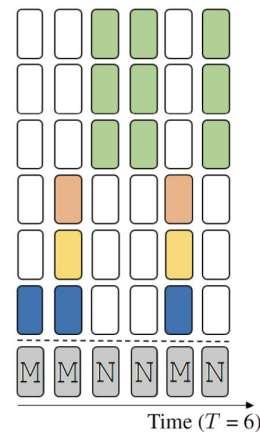
MIDI Meta Events: BPM, Time Signature, ...



(a) REMI representation



(b) Tokens grouped

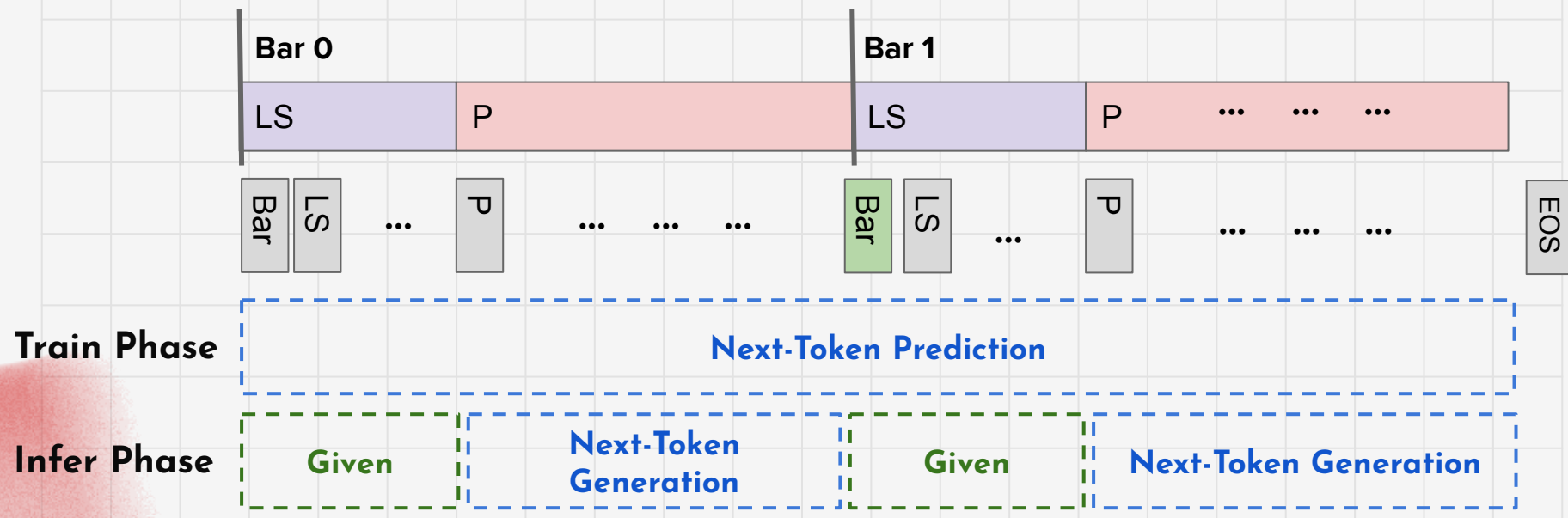


(c) Compound words

Symbolic to Audio to Domain - Example I

Conditional Generation, with decoder only (GPT-like) transformer

- Condition: Lead Sheet (L)
- Generation: Piano MIDI (P) - can be generalized to multi-track
- **T5 Prefix-LM** Mechanism



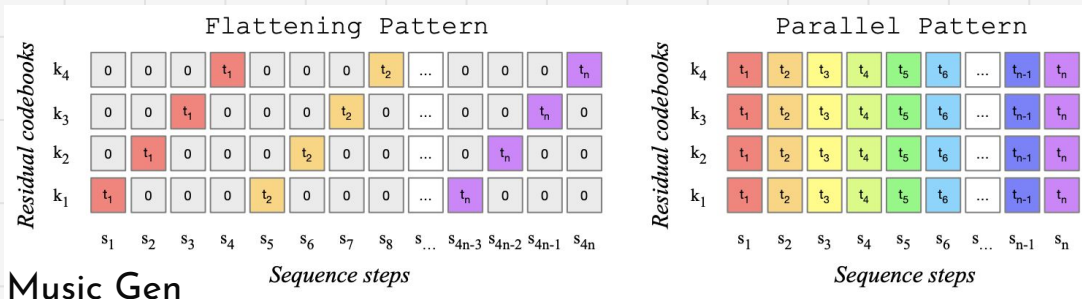
Symbolic to Audio to Domain - Example I

● Design Principle

- Token Length (Tokenization)
 - Length Compression
- Memory Complexity of Transformers
 - $O(N^2)$, N is seq len
 - Transformer-XL
 - Linear Transformer
- Sampling Policy
 - beam-search
 - Top-k, w/ temp
 - **Top-p**

CP Transformer

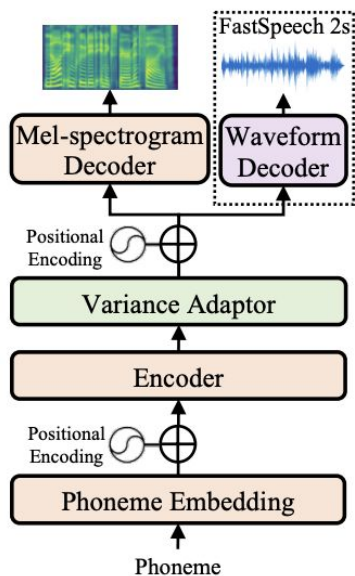
Task	Repre.	#words (T)	
		mean ($\pm$ std)	max
Conditional	REMI	6,432 ($\pm$ 1,689)	10,240
	CP	3,142 ($\pm$ 821)	5,120
Unconditional	REMI	4,873 ($\pm$ 1,311)	7,680
	CP	2,053 ($\pm$ 580)	3,584



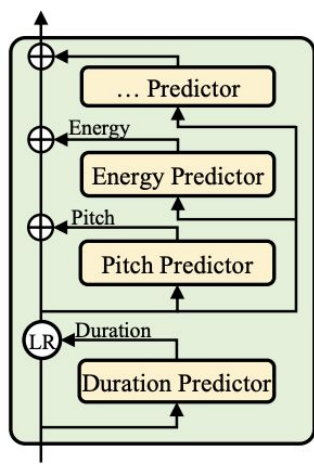
Symbolic to Audio to Domain - Example II

Singing Voice Synthesis = FastSpeech2 (modified) + Singing Vocoder

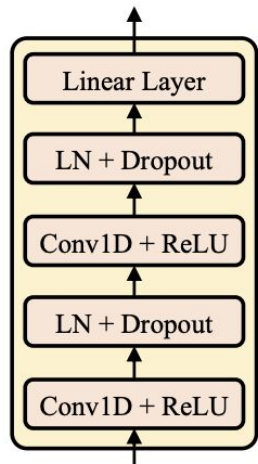
Original FastSpeech2



(a) FastSpeech 2

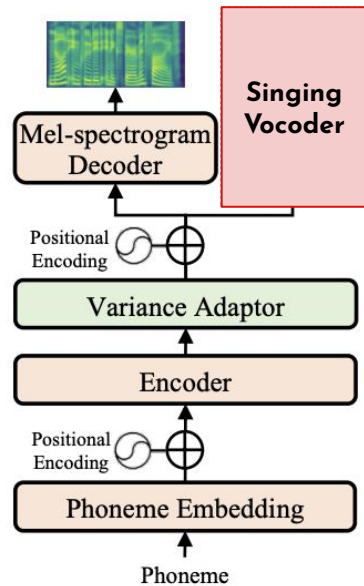


(b) Variance adaptor

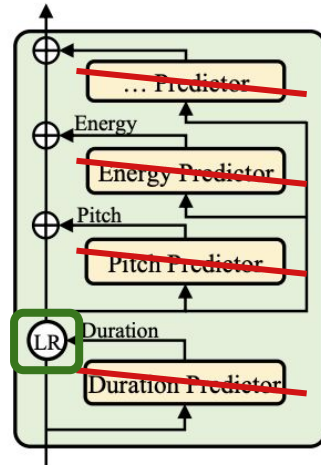


(c)
Duration/pitch/energy
predictor

Modified FastSpeech2 for Singing
Given: Duration/Pitch/Velocity(Energy)



(a) FastSpeech 2

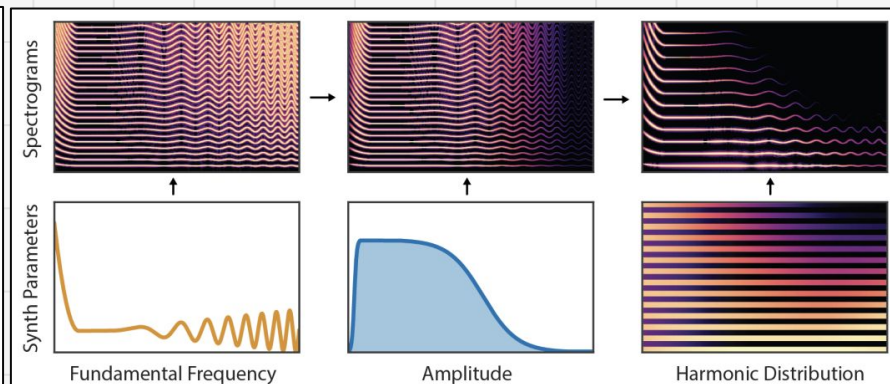
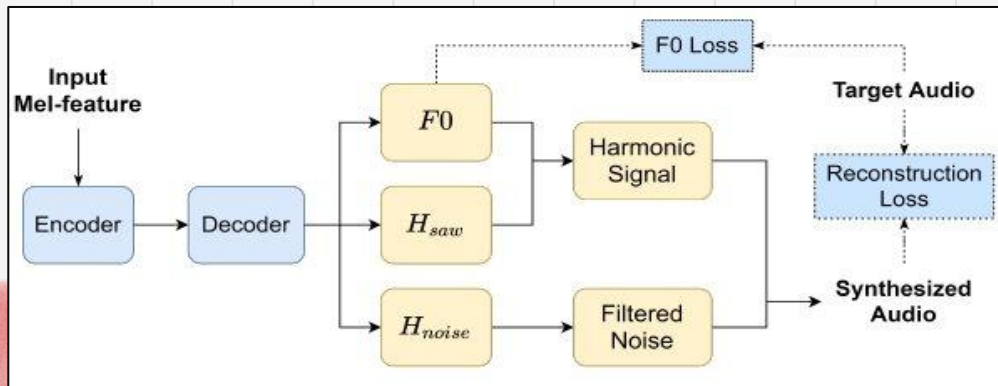
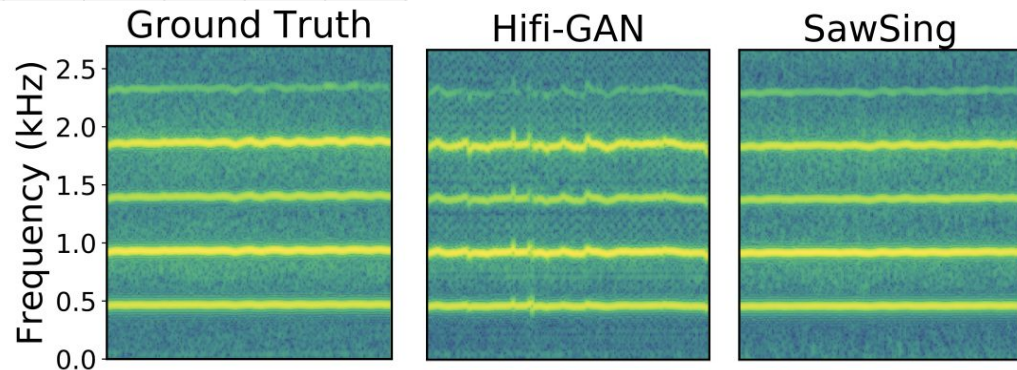


(b) Variance adaptor

Symbolic to Audio to Domain - Example II

DDSP Singing Vocoder (ISMIR'22)

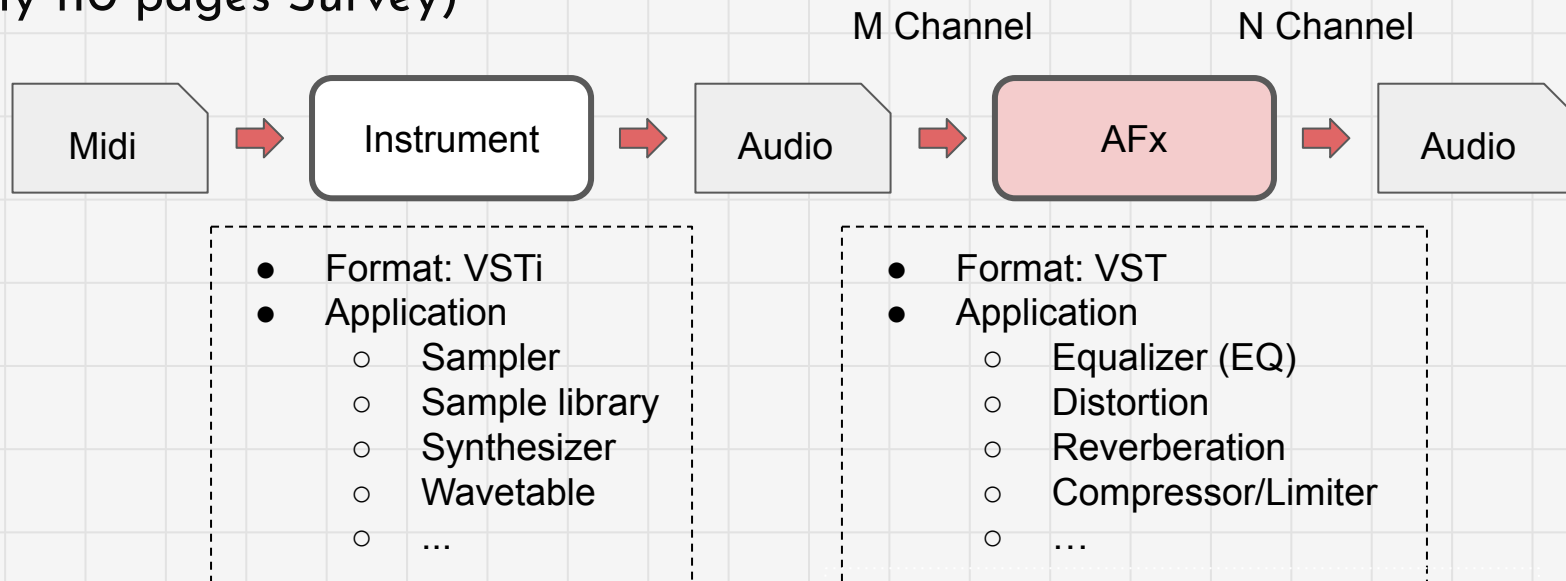
- DEMO
- NN-based Vocoder: slow
- No source signal input:
 - glitch in long utterance



Symbolic to Audio to Domain - Example III

Neural Audio Effect Modeling

(My 110 pages Survey)



x : input signal, M channel

y : output signal, N channel

c_g : global condition

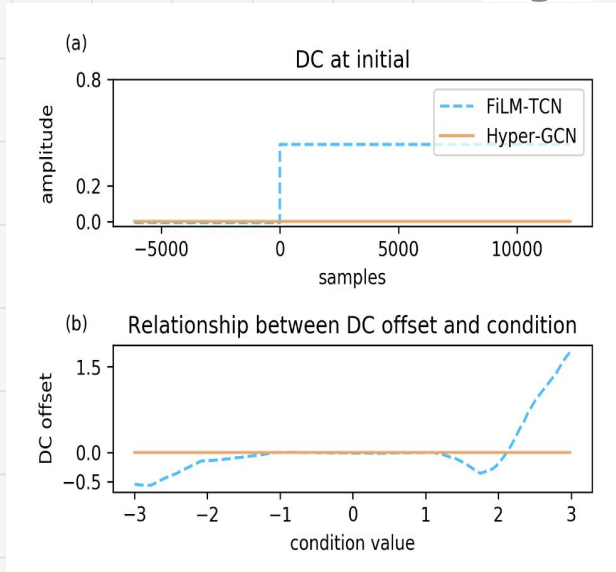
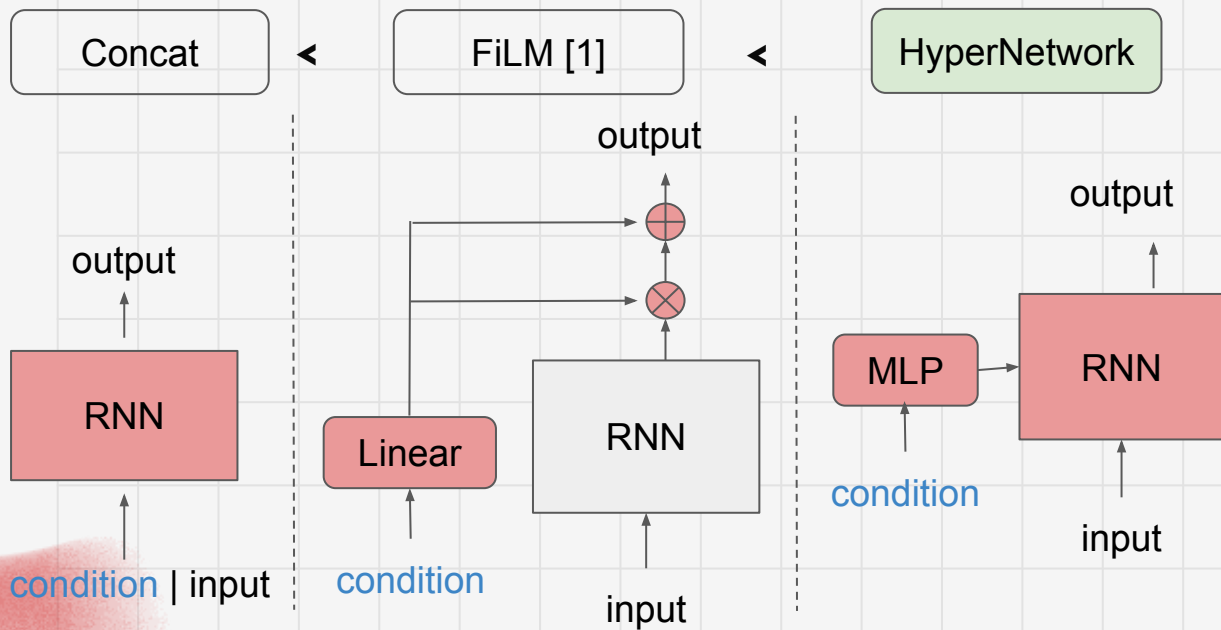
c_t : local condition

$$y = f(x, c_t, c_g)$$

Symbolic to Audio to Domain - Example III

Neural Audio Effect Modeling - DAFX'24 Oral

- Improve Quality and Solve DC Bias Issue

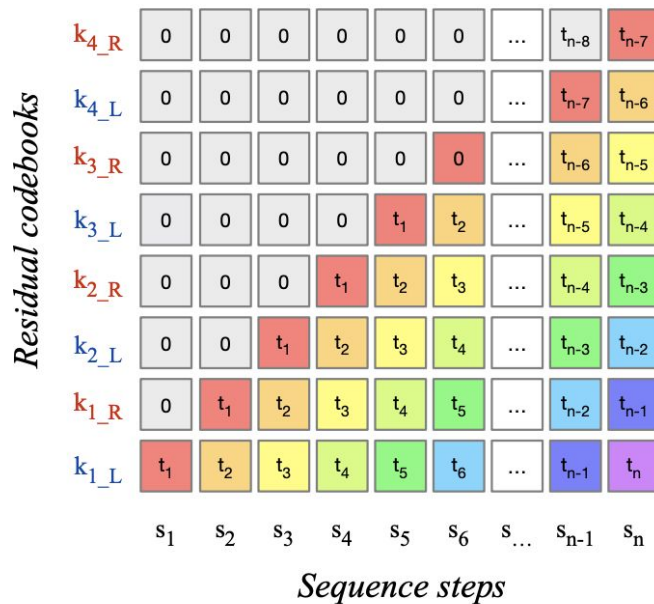


Symbolic to Audio to Domain - Example IV

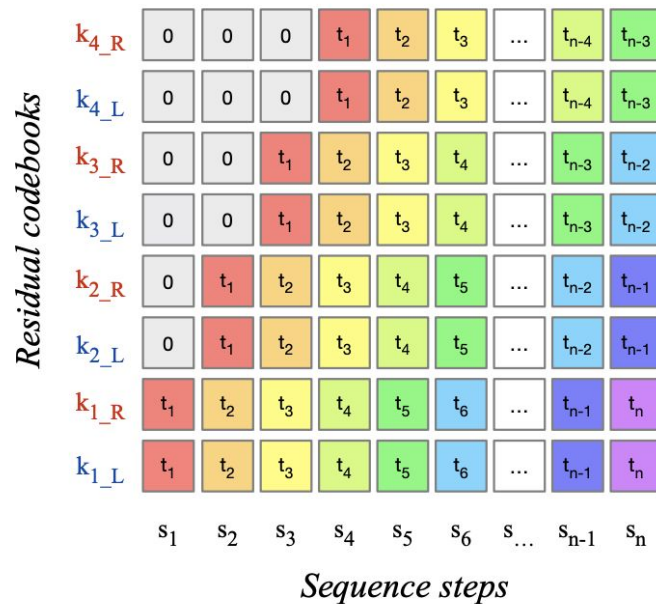
Text2Music with Temporal Controllation - MusicConGen (ISMIR'24)

- Fine-tune MusicGen (w/ melody) to control tempo & chord

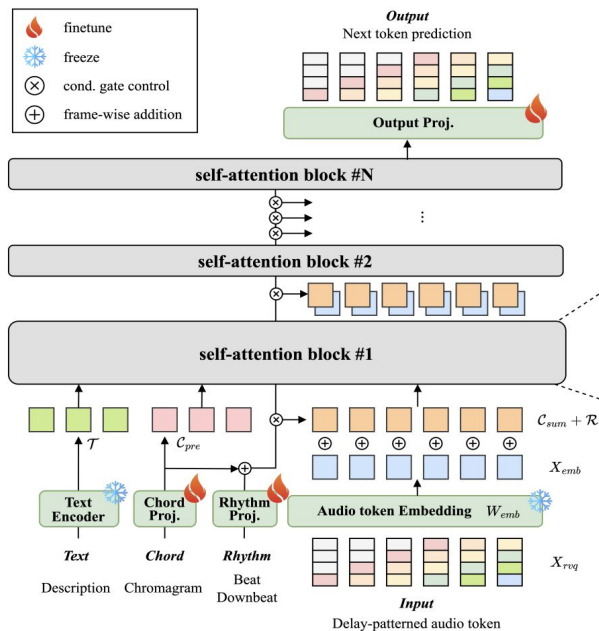
Stereo Delay Pattern



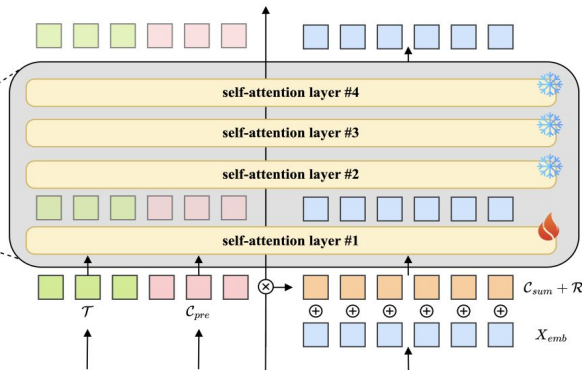
Stereo Partial Delay Pattern



Symbolic to Audio to Domain - Example IV



(a) MusiConGen model structure



(b) self-attention block

- MusicGen [1] =
 - RVQ [2] +
 - Flash-Attn [3]
- MusicConGen =
 - Module Reuse
 - Fine-tuning
- Fine-tune on
 - Single RTX3090
 - Inhouse Data

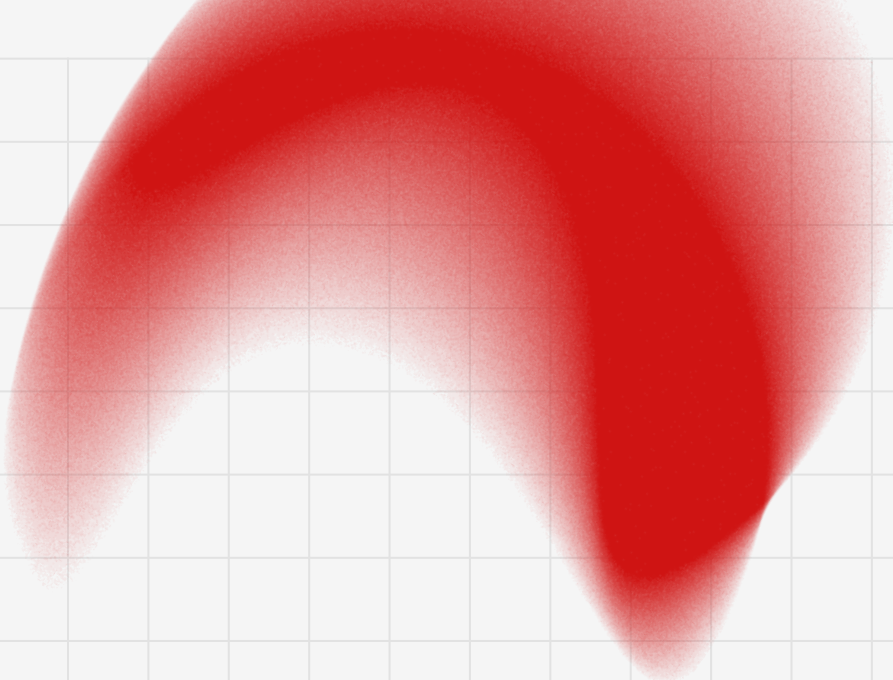
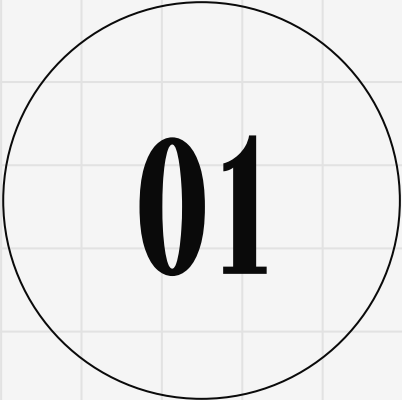
DEMO

Reference Chords																																			
Generated Sample's Chords																																			
	0.410.821.221.632.032.442.833.253.664.064.464.875.265.686.076.496.97.317.78.128.528.939.319.7310.1410.5510.9411.3611.7512.1712.5612.9813.3913.7914.1914.615.015.4115.816.2216.6317.0417.4317.8518.2418.6519.0419.4719.8820.2820.6821.0921.4921.922.2922.7123.1223.5323.9224.3424.7725.1425.5325.9526.3626.7727.1627.5828.028.3928.7729.1629.6																																		
description	A laid-back blues shuffle with a relaxed tempo, warm guitar tones, and a comfortable groove, perfect for a slow dance or a night in. Instruments: electric guitar, bass, drums.							A smooth acid jazz track with a laid-back groove, silky electric piano, and a cool bass, providing a modern take on jazz. Instruments: electric piano, bass, drums.							A classic rock n' roll tune with catchy guitar riffs, driving drums, and a pulsating bass line, reminiscent of the golden era of rock. Instruments: electric guitar, bass, drums.							A high-energy funk tune with slap bass, rhythmic guitar riffs, and a tight horn section, guaranteed to get you grooving. Instruments: bass, guitar, trumpet, saxophone, drums.							A heavy metal onslaught with double kick drum madness, aggressive guitar riffs, and an unrelenting bass, embodying the spirit of metal. Instruments: electric guitar, bass guitar, drums.						
Sample 001	▶ 0:00 / 0:30 — 🔊 ⋮							▶ 0:00 / 0:30 — 🔊 ⋮							▶ 0:00 / 0:30 — 🔊 ⋮							▶ 0:00 / 0:30 — 🔊 ⋮							▶ 0:00 / 0:30 — 🔊 ⋮						

- [1] Simple and Controllable Music Generation (Neurips'23)
- [2] High-Fidelity Audio Compression with Improved RVQGAN (Neurips'23)
- [3] FlashAttention

Thank you

My Paper Reading Notes



01

LLM & FM

Large Language Model (LLM) and
Foundation Model (FM)

LLM & FM – Definition

- What is **Language Model (LM)**?

$$\begin{aligned} P_{(w_1, w_2, \dots, w_n)} &= p(w_1)p(w_2|w_1)p(w_3|w_1, w_2)\dots p(w_n|w_1, w_2, \dots, w_{n-1}) \\ &= \prod_{i=1}^n p(w_i|w_1, \dots, w_{i-1}) \end{aligned} \quad (1)$$

- What is **Large Language Model (LLM)**?
 - It's LM trained on large corpus with large amount of parameters (Billion/Trillion).
 - ChaptGPT, LLama
- What is **Foundation Model (FM)**?
 - It's a broader concept including LLMs
 - Multimodal data, including images, audio, video, and text.
 - In a Paradigm like {Petrained, Fine-Tuning}
 - GPT, CLIP, CLAP, BERT

LLM & FM – Examples

Three families:

- **BERT-like (Transformer Encoder)**
- **GPT-like (Transformer Decoder)**
- **CLIP-like (Contrastive Learning)**

Two Topics:

- **How to Fine-Tuning?**
- **Hallucination**

BERT-like

- **Training Goals**

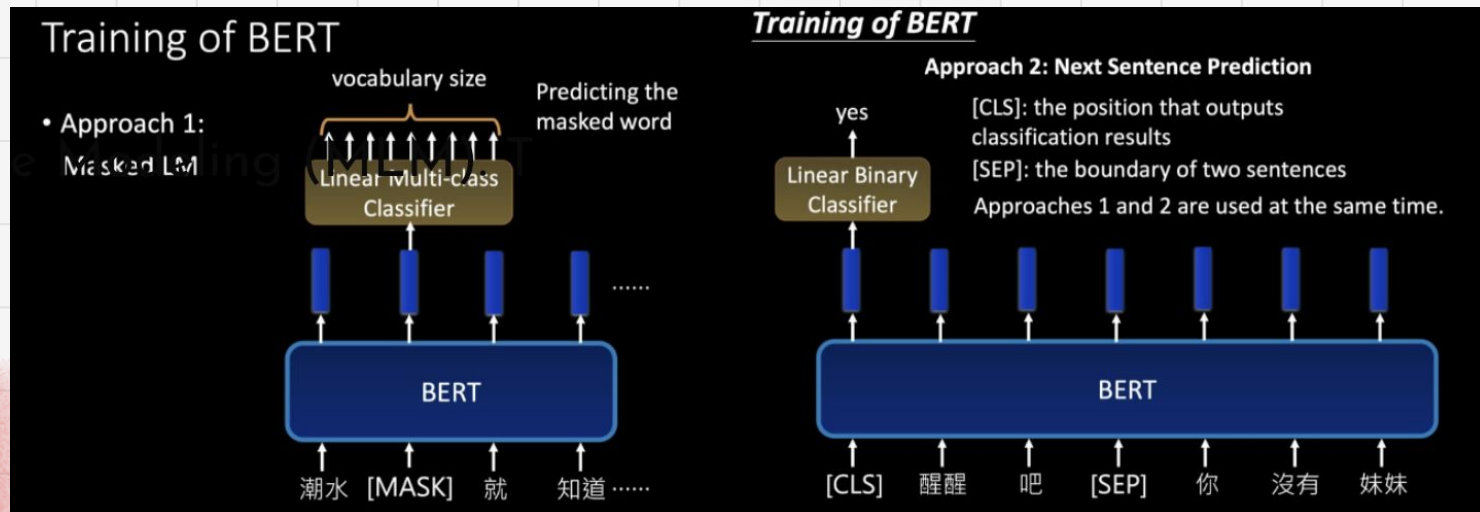
- **Masked Language Modeling (MLM).**
- **Next Sentence Prediction (NSP).**

- **Applications**

- Feats for Downstream task

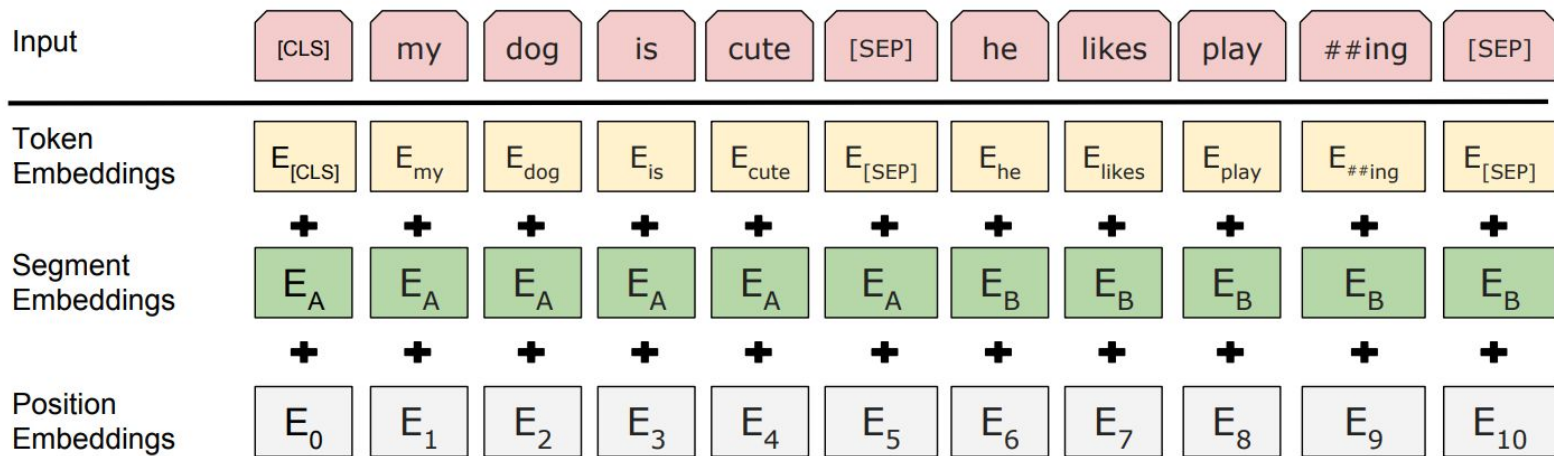
- **Difficulties for other domain**

- Transformers work on discrete tokens
- How to discretize continuous feats?
 - Spectrogram, Imagem ...



BERT-like

- **Backbone Model - Transformer Encoder**
 - No causal mask
 - Bidirectional
 - Non-autoregressive model



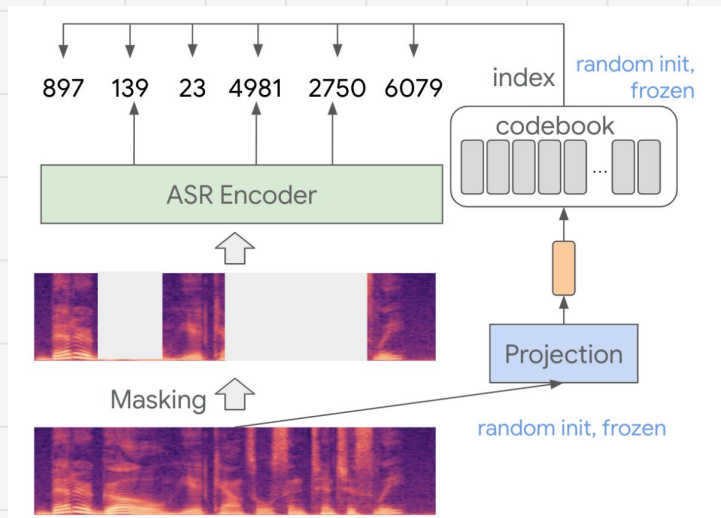
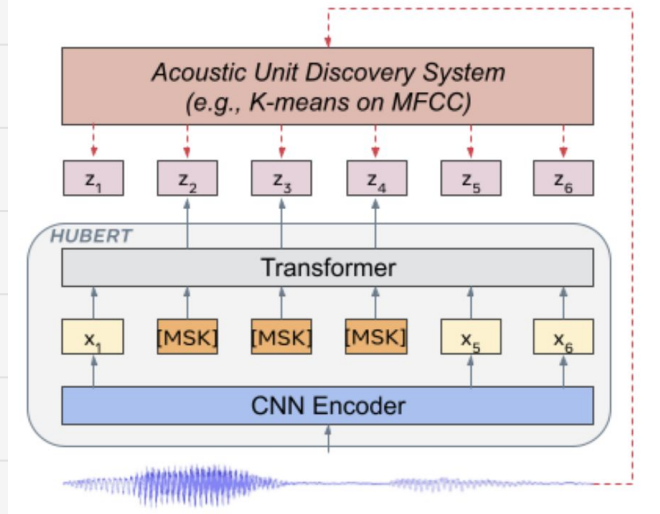
BERT-like

- **Difficulties for other domain**

- Transformers work on discrete tokens
- How to discretize continuous feats?
 - Spectrogram, Image, ...

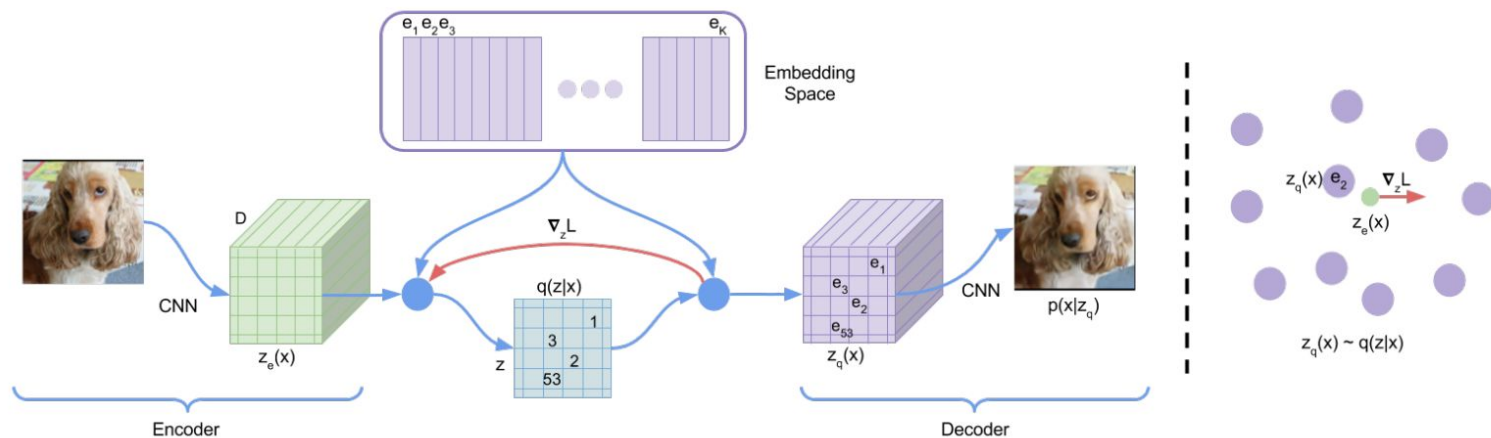
- **Examples on Audio**

- Wav2Vec, HuBERT, Best RQ.
- How to discretize audio waveform?
 - Hubert: K-means Clustering
 - Best RQ: Vector Quantizer



BERT-like

- Discretization and Quantization
 - VQVAE (for image)



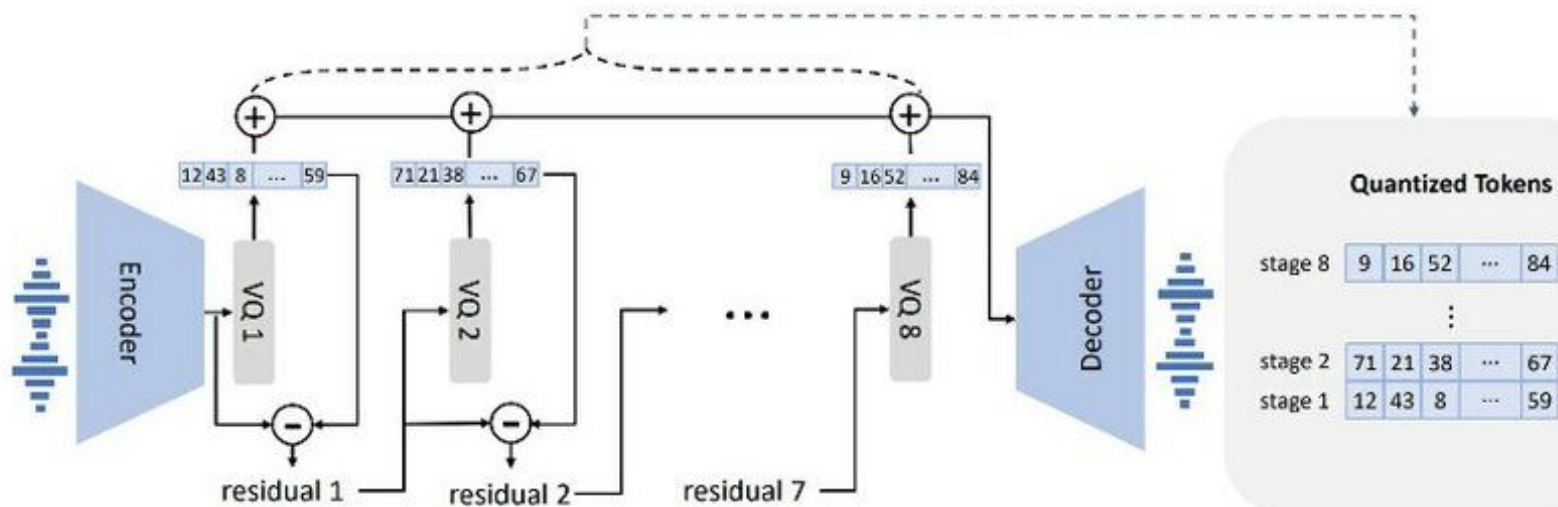
$$L = \log p(x|z_q(x)) + \|\text{sg}[z_e(x)] - e\|_2^2 + \beta \|z_e(x) - \text{sg}[e]\|_2^2, \quad (3)$$

where sg stands for the stopgradient operator that is defined as identity at forward computation time and has zero partial derivatives, thus effectively constraining its operand to be a non-updated constant.

The decoder optimises the first loss term only, the encoder optimises the first and the last loss terms, and the embeddings are optimised by the middle loss term. We found the resulting algorithm to be

BERT-like

- **Improved Version - RVQ**
 - SoundStream (from google)
 - Encodec (from Meta)



GPT-like

- **Training Goals**
 - Next Token Prediction
- **Backbone Model:**
 - **Transformer Decoder**
 - Training
 - Causal Masked
 - Inference
 - Auto-regressive
 - Sampling
- **Applications**
 - "LMs are Few-Shot Learners"
 - Prompt Interaction

The three settings we explore for in-context learning

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée ←
4 plush girafe => girafe peluche ←
5 cheese => ..... ← prompt
```

Traditional fine-tuning (not used for GPT-3)

Fine-tuning

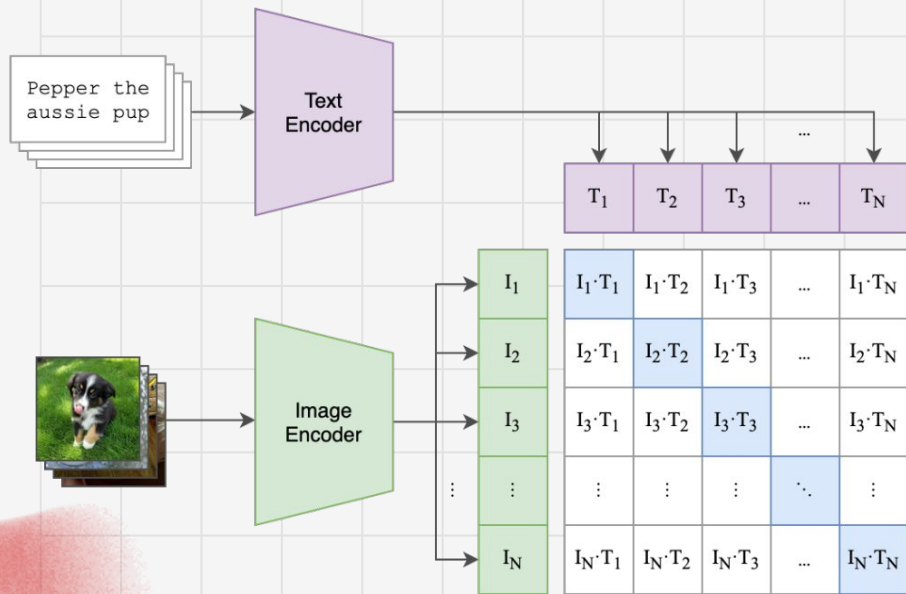
The model is trained via repeated gradient updates using large corpus of example tasks.



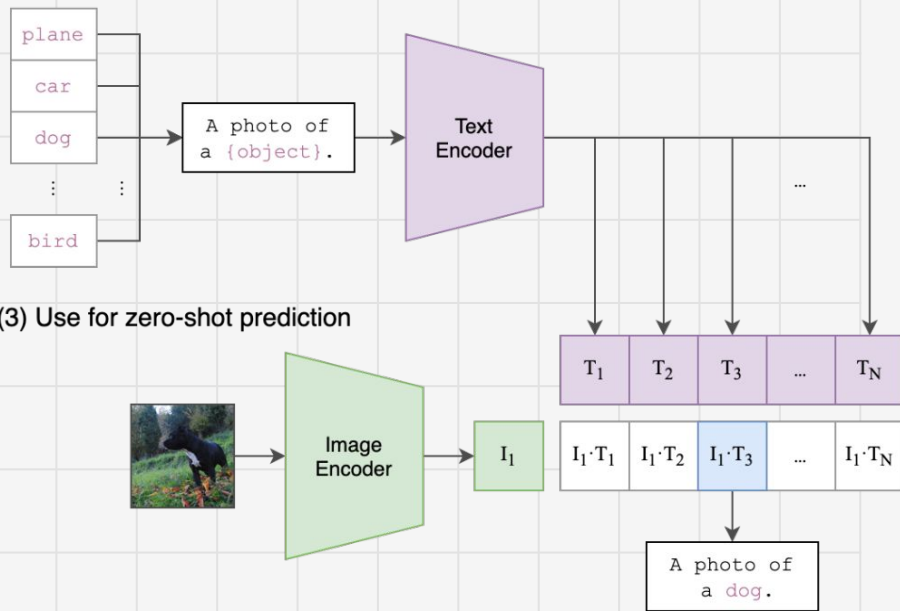
CLIP-like

CLAP: Image x Text

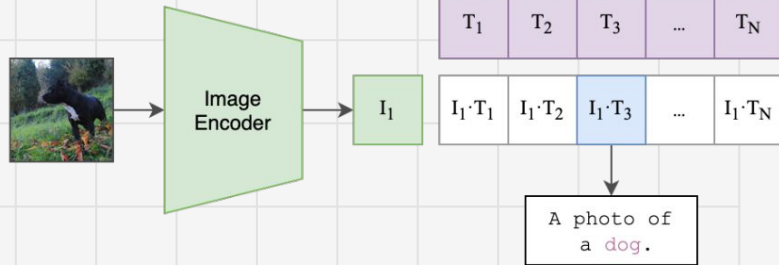
(1) Contrastive pre-training



(2) Create dataset classifier from label text

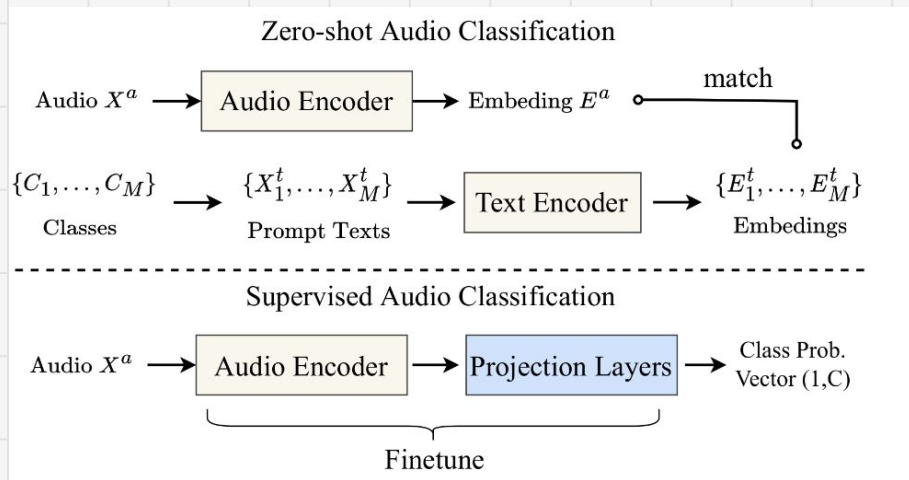
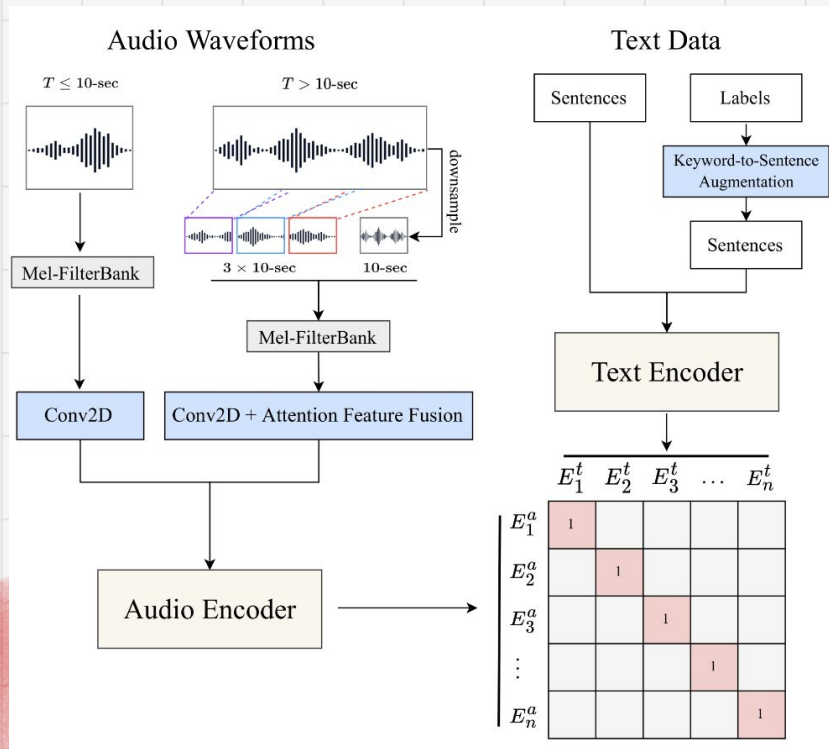


(3) Use for zero-shot prediction



CLIP-like

CLAP: Audio x Audio



How to Fine-tuning

- **Supervised Fine-Tuning (SFT)**
 - Use small and clean high quality data
 - Freeze part of trainable models, small learning rate
 - **Cons:** Gradient Update required
- **Reinforcement Learning with Human Feedback (RLHF)**
 - Similar to SFT, different rewarding policy
 - **Cons:** human annotation -> resource-intensive
- **Prompt Engineering**
 - **Zero-shot, One-shot, Few-shot**
 - **Pros:** no Gradient update

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
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3 cheese => ..... ← prompt
```

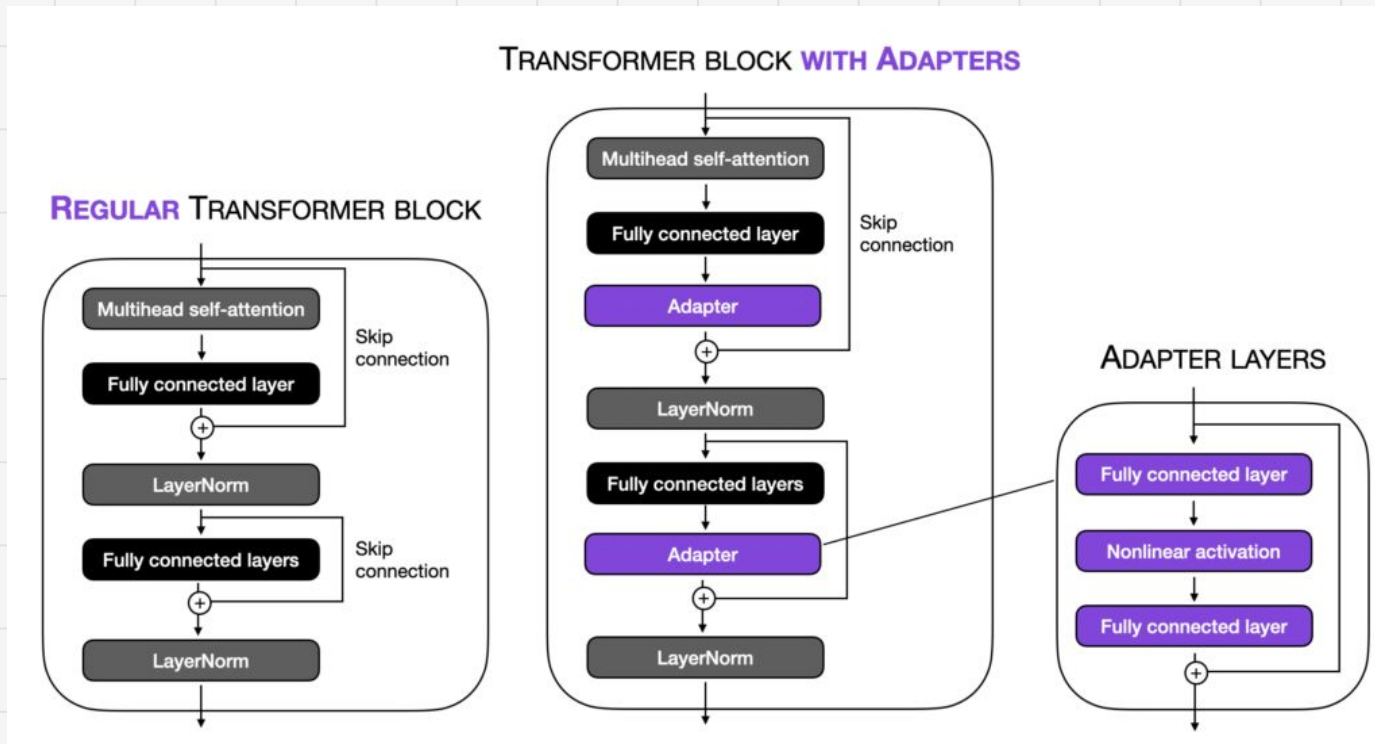
Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée ←
4 plush girafe => girafe peluche ←
5 cheese => ..... ← prompt
```

How to Fine-tuning

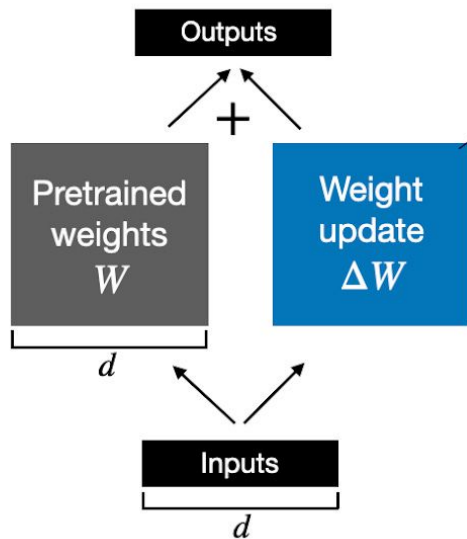
- **Adapter Layers (LLaMA-Adapte)**



How to Fine-tuning

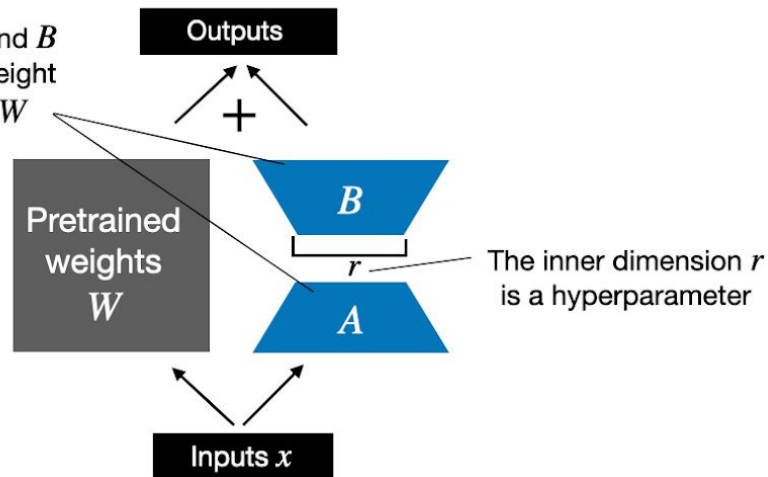
- LoRA

Weight update in **regular finetuning**



LoRA matrices A and B approximate the weight update matrix ΔW

Weight update in **LoRA**



Hallucination

The model generates **fake or fabricated information** but is **delivered confidently**.

The generated content is **not coherent to reality**.

- **Why**

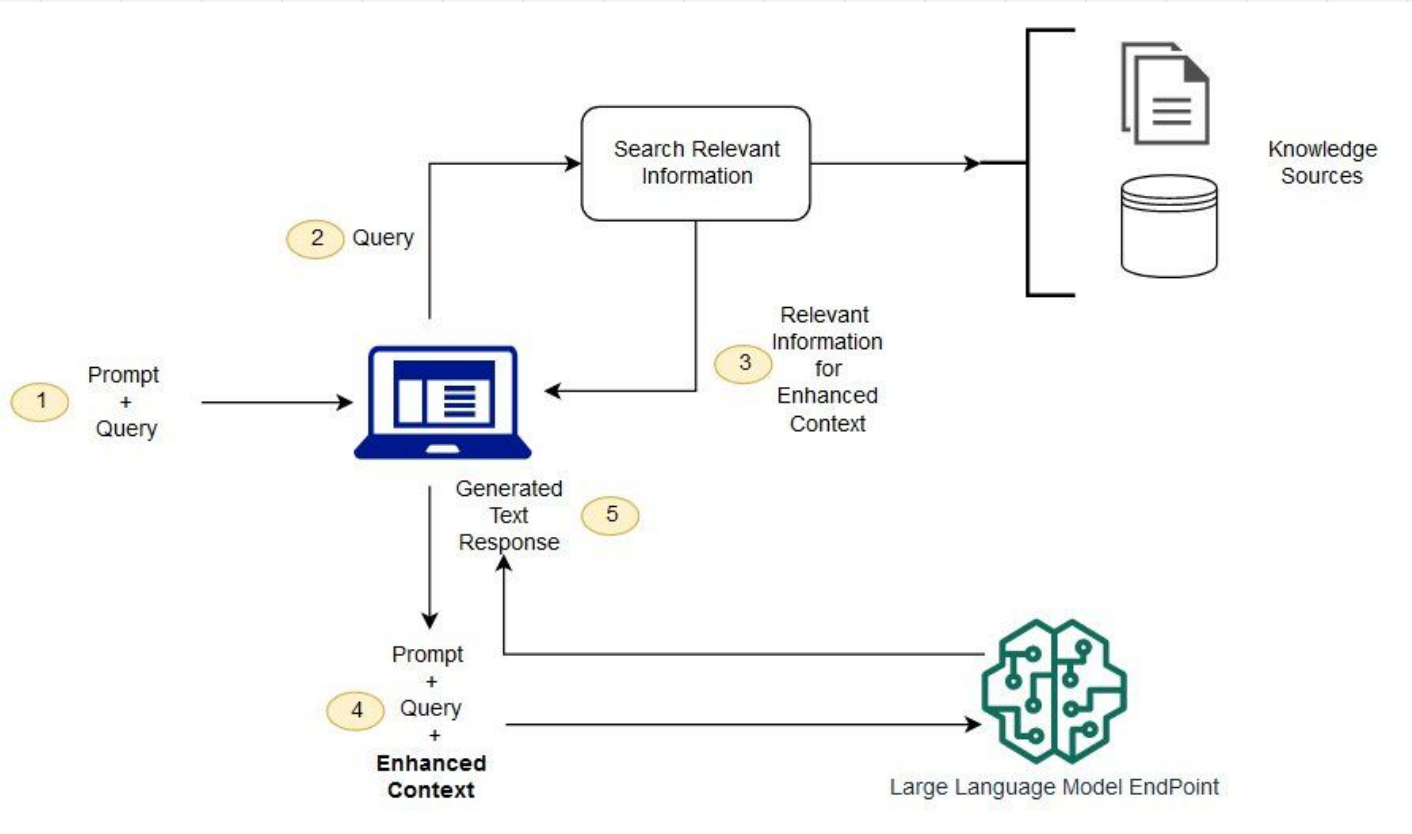
- Training on Noisy/Biased/Inaccurate/Outdated Data
- Training Objectives
 - Modern ML is more like a Pattern Recognition/probabilistic model.
 - It's not based on reasoning and not interact with real-world
- Context Length
 - While training, the sequence length of training samples is limited
 - While generating long content, the model tends to forget the past

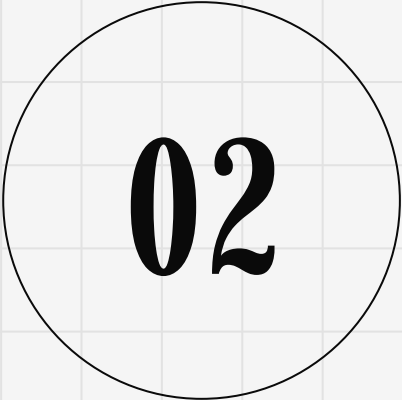
- **Solution**

- Prompt Engineering
- Fine-Tuning
- **Integrate with external data - RAG**

Hallucination - RAG

RAG (Retriever-Augmented Generation)





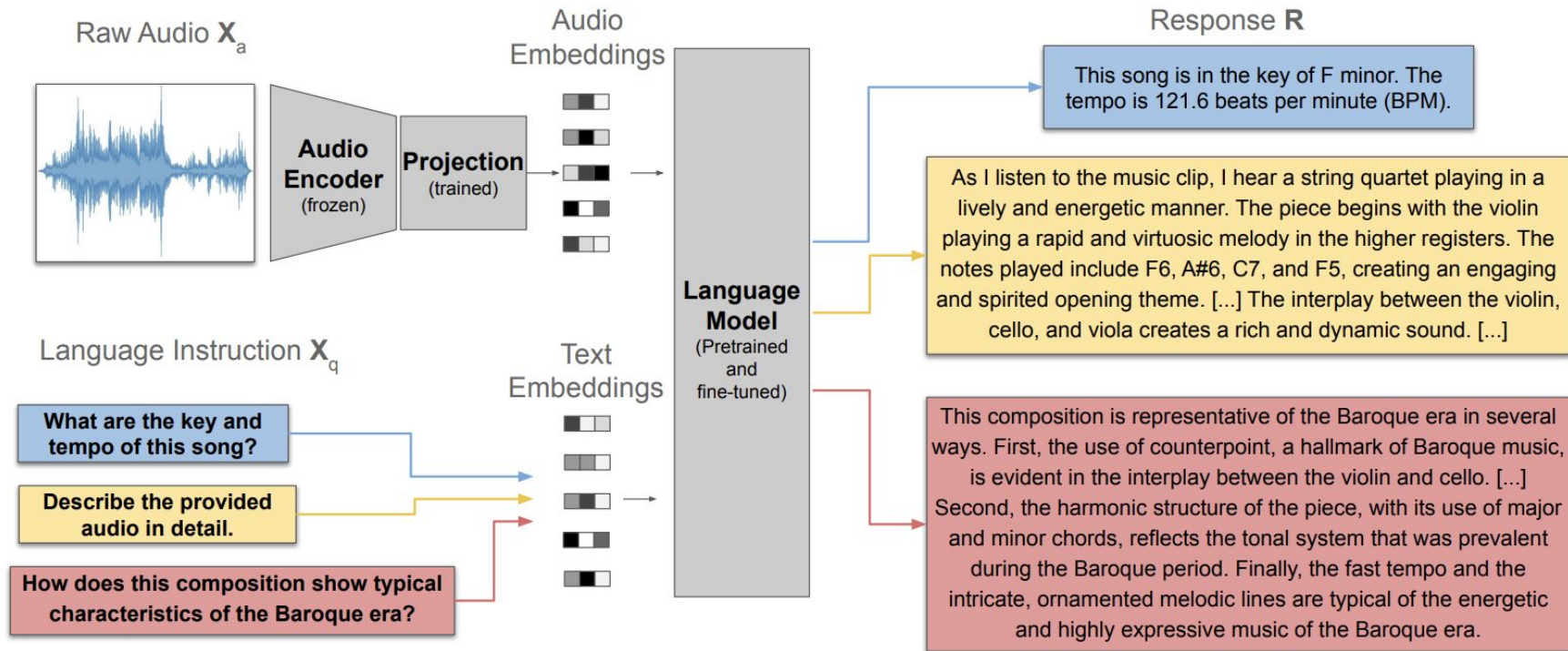
02

LLM & FM on Music

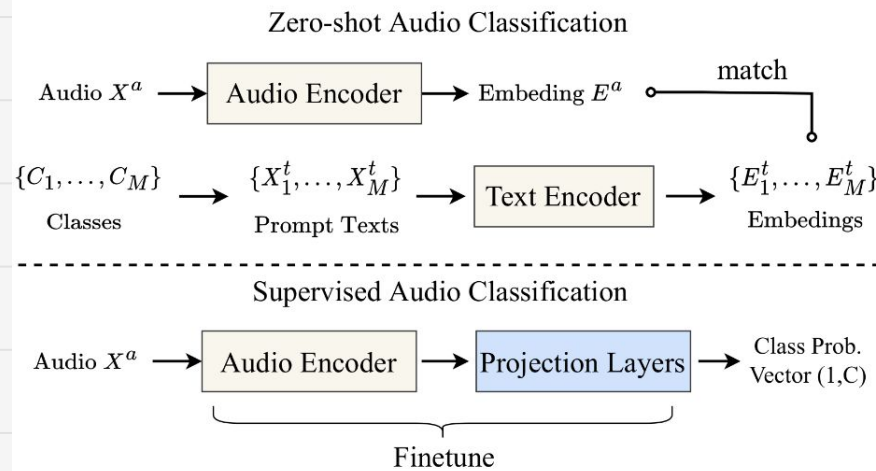
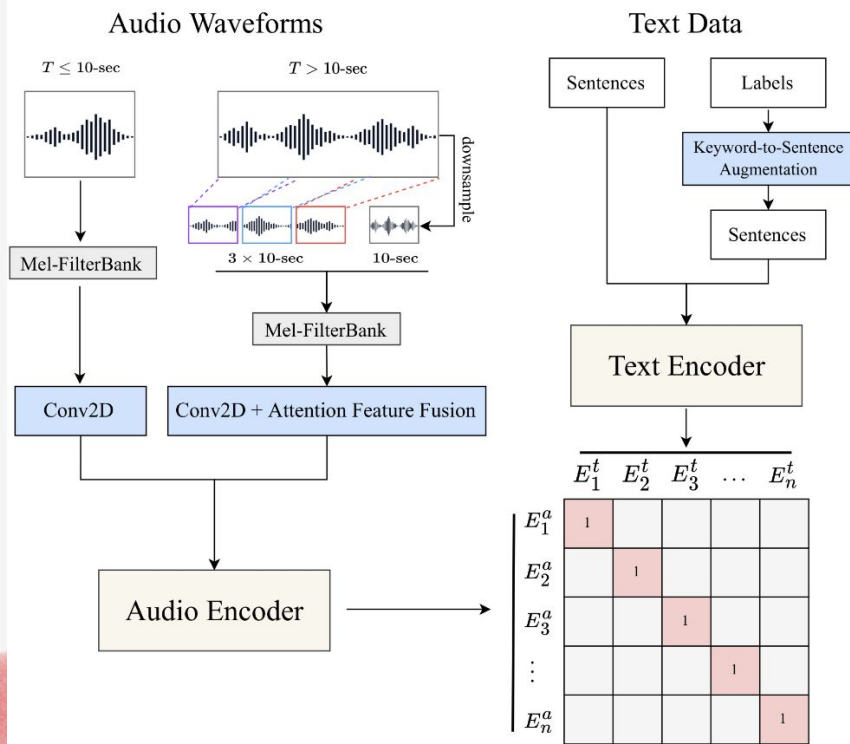
LLM on Music

- **LLark**
- **CLAP**
- **MERT**
- **MusicGen**
- **MT3**
- **Foundation Model Survey**

LLark (from Spotify)



CLAP (from LAION-AI)



$$\text{CLIPScore}(I, C) = \max(100 * \cos(E_I, E_C), 0)$$

Contrastive Learning

```
# image_encoder - ResNet or Vision Transformer
# text_encoder - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T) #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss = (loss_i + loss_t)/2
```

```
import torch
import torch.nn as nn
import torch.nn.functional as F
import numpy as np

from simple_clip.utils import get_feature_size

def contrastive_loss(logits):
    targets = torch.arange(logits.size(0)).to(logits.device)
    loss_images = F.cross_entropy(logits, targets)
    loss_texts = F.cross_entropy(logits.t(), targets)
    return (loss_images + loss_texts) / 2

def siglip_loss(logits):
    n = logits.size(0)
    # -1 for off-diagonals and 1 for diagonals
    labels = 2 * torch.eye(n, device=logits.device) - 1
    # pairwise sigmoid loss
    return -torch.sum(F.logsigmoid(labels * logits)) / n

class CLIP(torch.nn.Module):
    def __init__(self,
                 image_encoder,
                 text_encoder,
                 image_mlp_dim=False,
                 text_mlp_dim=768,
                 proj_dim=256,
                 init_tau=np.log(1.0),
                 init_b=0):
        super(CLIP, self).__init__()

        if not image_mlp_dim:
            image_mlp_dim = get_feature_size(image_encoder)

        self.image_encoder = image_encoder
        self.text_encoder = text_encoder

        self.image_projection = torch.nn.Sequential(
            torch.nn.Linear(image_mlp_dim, image_mlp_dim, bias=False),
            torch.nn.ReLU(),
            torch.nn.Linear(image_mlp_dim, proj_dim, bias=False))

        self.text_projection = torch.nn.Sequential(
            torch.nn.Linear(text_mlp_dim, text_mlp_dim, bias=False),
            torch.nn.ReLU(),
            torch.nn.Linear(text_mlp_dim, proj_dim, bias=False))

        self.t_prime = nn.Parameter(torch.ones(1) * init_tau)
        self.b = nn.Parameter(torch.ones(1) * init_b)

    def forward(self, image, input_ids, attention_mask):
        image_features = self.extract_image_features(image)
        text_features = self.extract_text_features(input_ids, attention_mask)
        image_features = F.normalize(image_features, p=2, dim=-1)
        text_features = F.normalize(text_features, p=2, dim=-1)
        return image_features @ text_features.t() * self.t_prime.exp() + self.b

    def extract_image_features(self, images):
        image_features = self.image_encoder(images)
        return self.image_projection(image_features)

    def extract_text_features(self, input_ids, attention_mask):
        text_features = self.text_encoder(input_ids, attention_mask)
        return self.text_projection(text_features)
```

MERT (from LAION-AI)

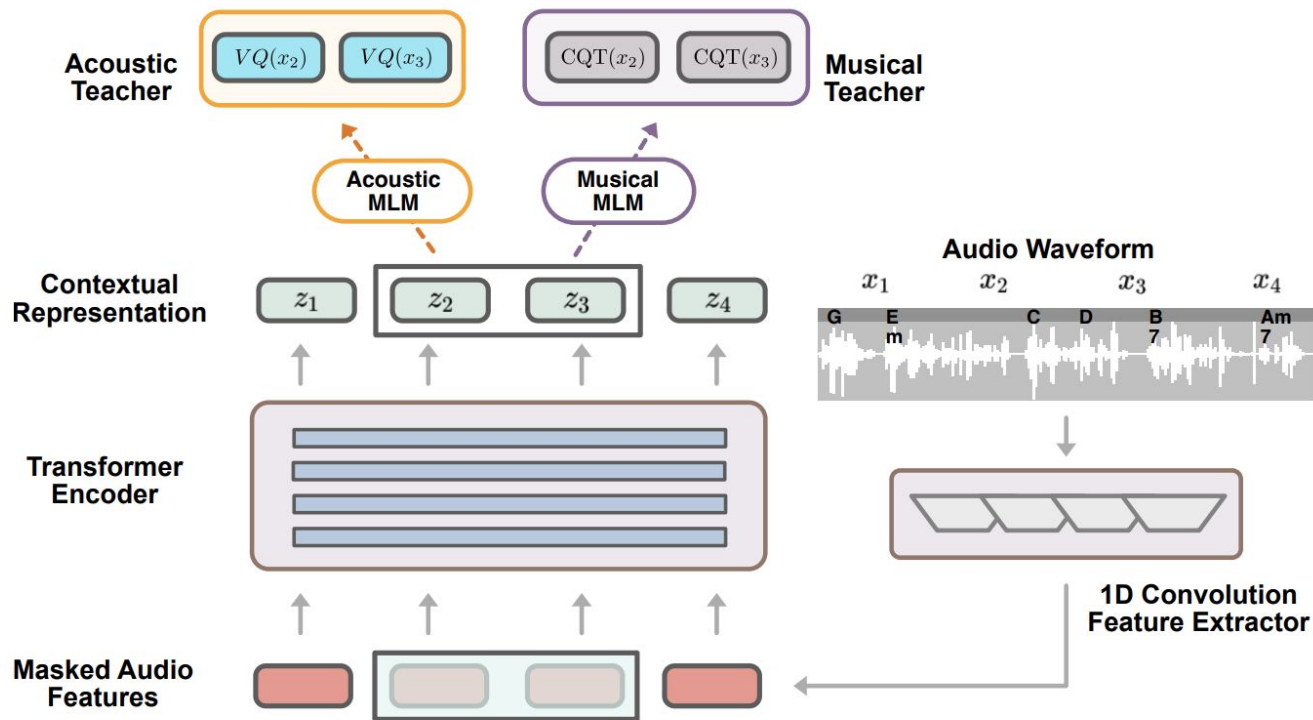
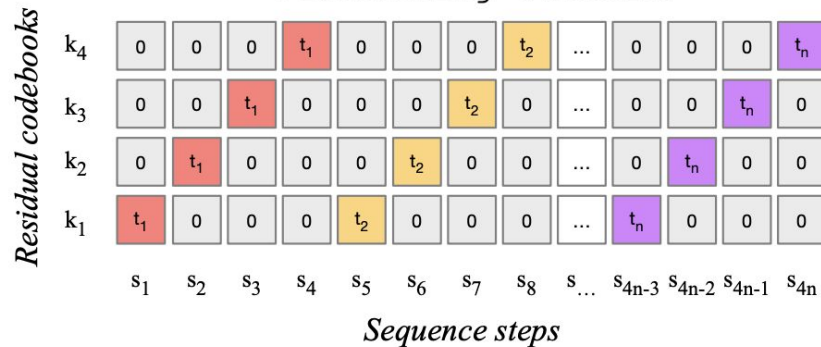


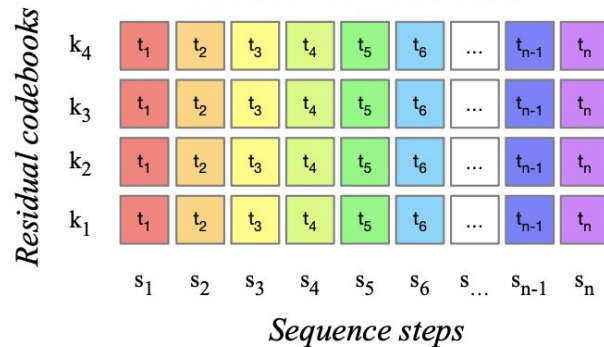
Figure 1: Illustration of the MERT Pre-training Framework.

MusicGen (from Meta)

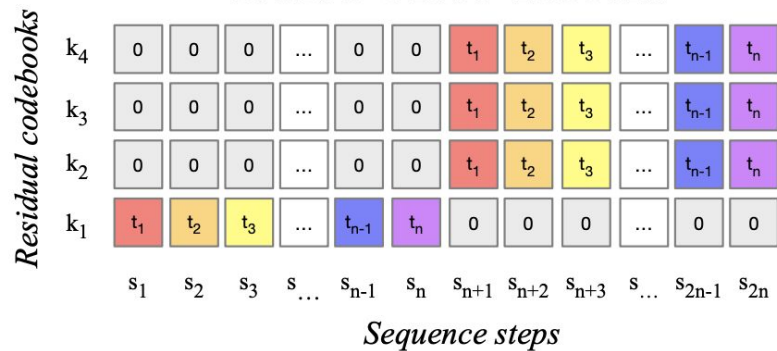
Flattening Pattern



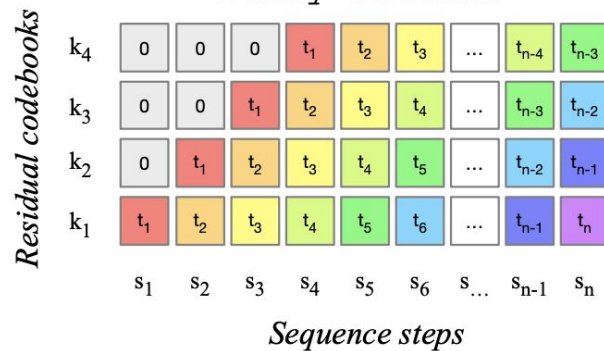
Parallel Pattern



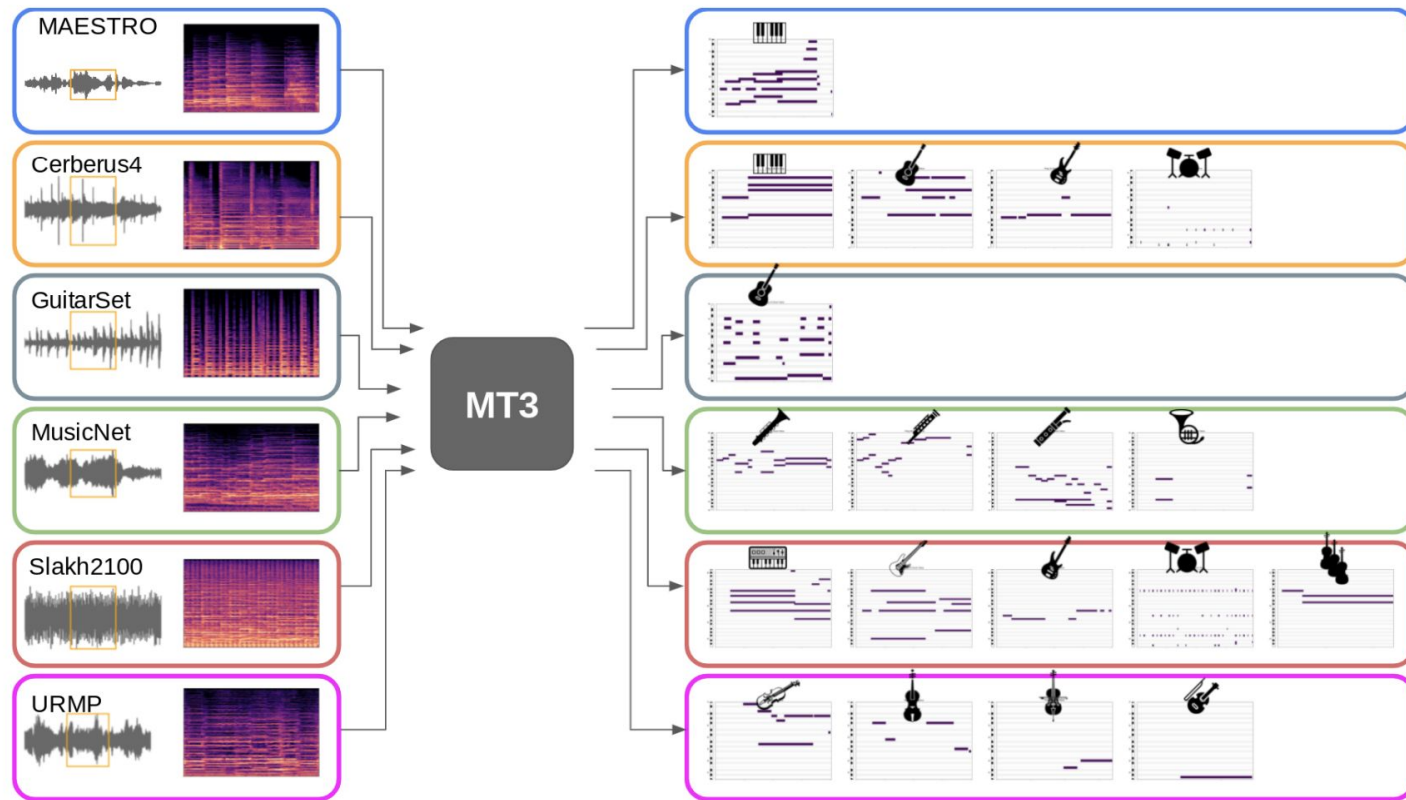
Coarse First Pattern



Delay Pattern

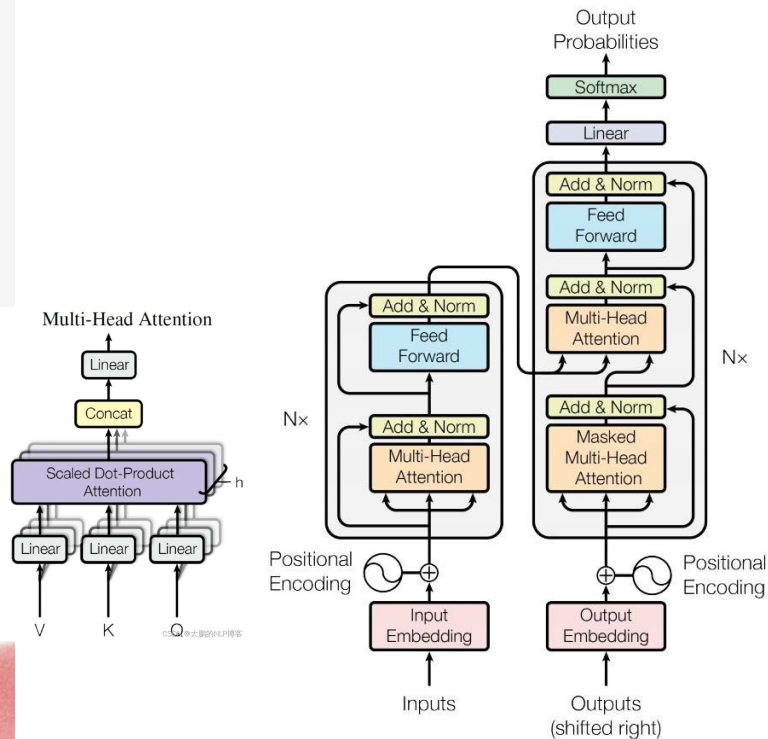


MT3 (from Google)



Idea: audio + prompt (instrument) -> corresponding score (midi, tab)

Transformer



Docoder Only: Masking Policy

[illegible]

Transformer

$$f_{t:t \in \{q,k,v\}}(\mathbf{x}_i, i) := \mathbf{W}_{t:t \in \{q,k,v\}}(\mathbf{x}_i + \mathbf{p}_i),$$

$$\begin{cases} \mathbf{p}_{i,2t} &= \sin(k/10000^{2t/d}) \\ \mathbf{p}_{i,2t+1} &= \cos(k/10000^{2t/d}) \end{cases}$$

Absolute

$$f_q(\mathbf{x}_m) := \mathbf{W}_q \mathbf{x}_m$$

$$\begin{aligned} f_k(\mathbf{x}_n, n) &:= \mathbf{W}_k(\mathbf{x}_n - \tilde{\mathbf{p}}_r^k) \\ f_v(\mathbf{x}_n, n) &:= \mathbf{W}_v(\mathbf{x}_n - \tilde{\mathbf{p}}_r^v) \end{aligned}$$

Relative

trainable

Positional Embedding

- Absolute
- Relative (pairwise)
- Rope = Abs + Rel
- Alibi = Rope + Extrapolatio

absolute

for pair-wise

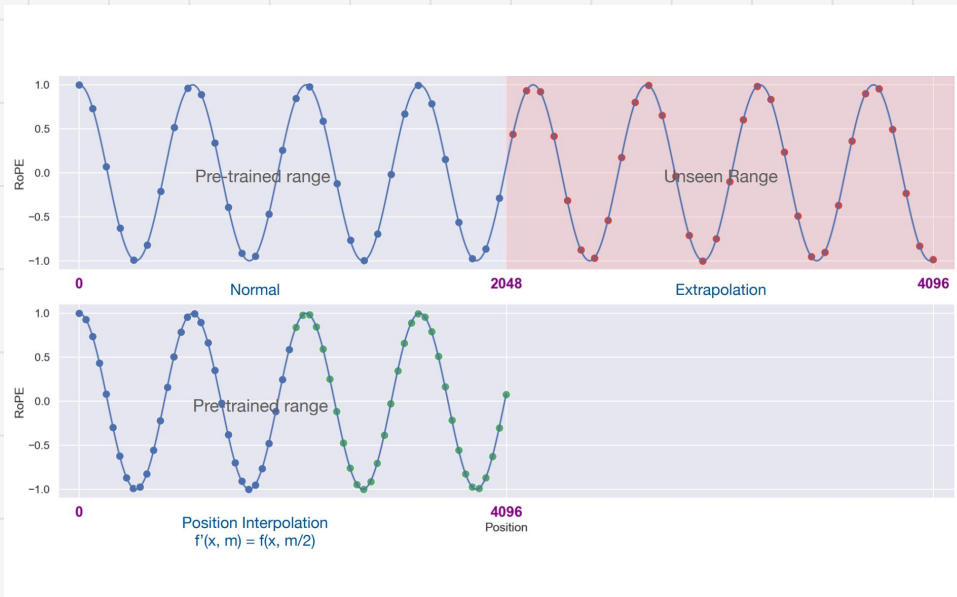
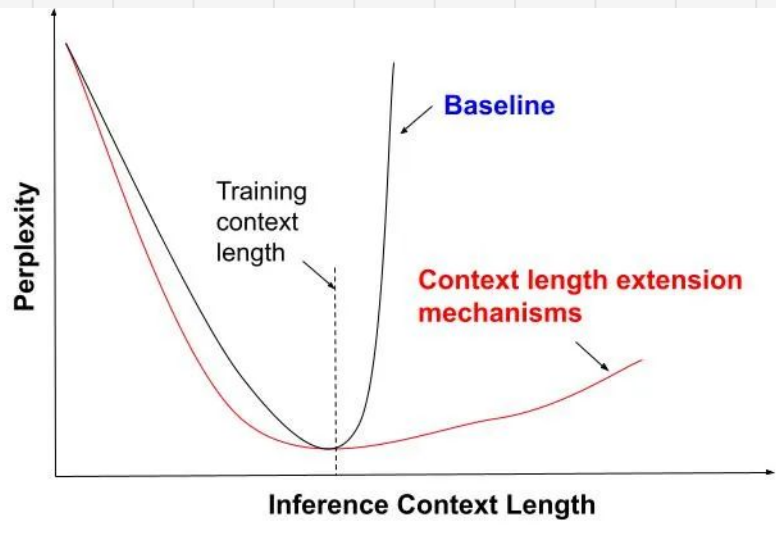
for pair-wise and absolute

for context extending

(ex: Vanilla Transformer)

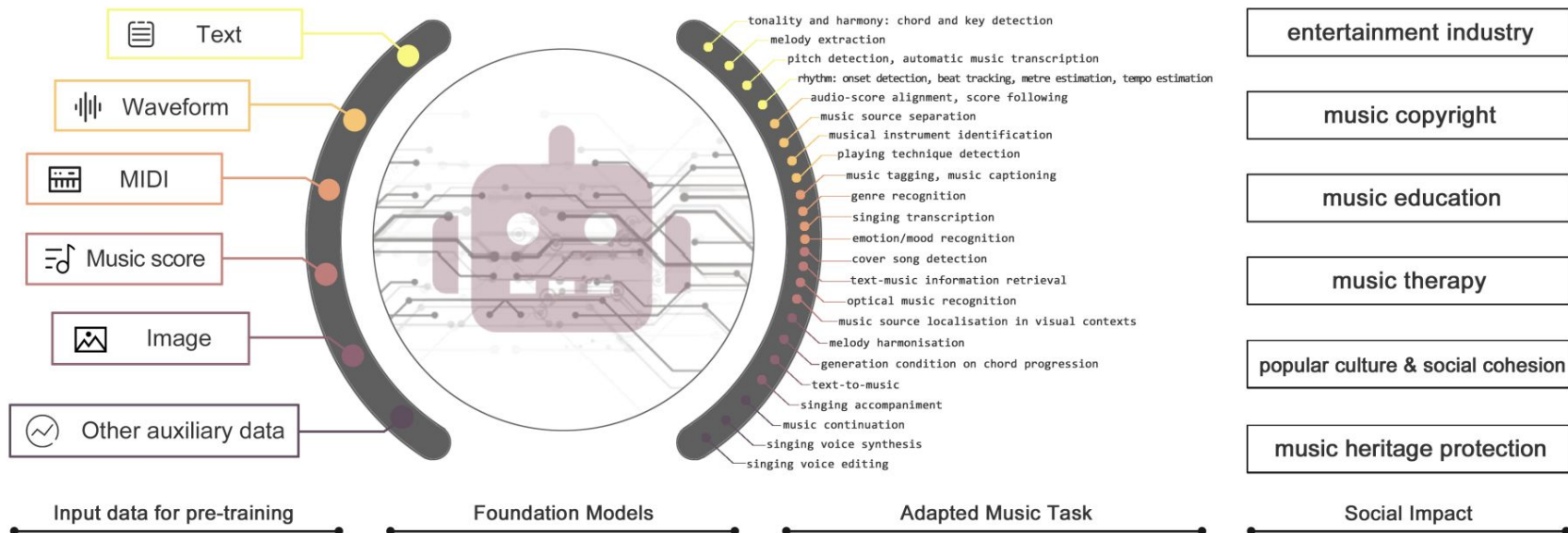
(ex: Transformer-XL)

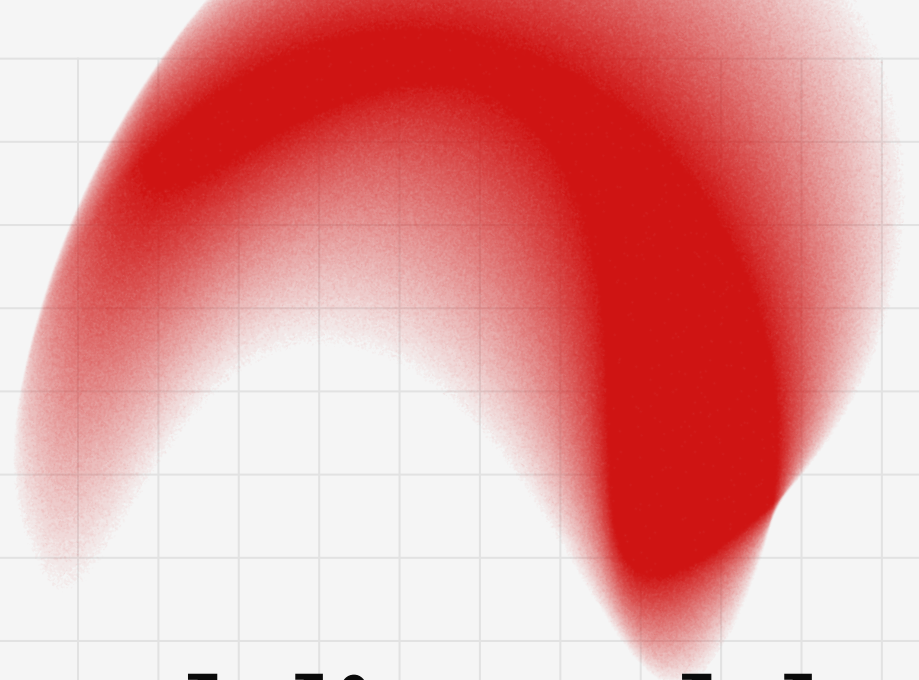
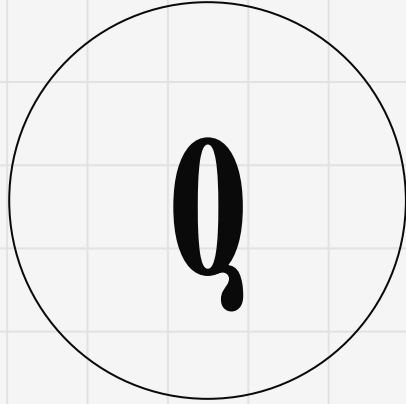
(ex: LLama, ChatGLM)



Foundation Model Survey (latest)

[Link](#) - [arxiv.2408](#)





Design Multi-Modality Model

SonyCSL

Q: Designing Multi-Modal Models Combining **Audio** and **Symbolic** Representations

Goal : Propose a framework for a generative model that simultaneously learns from both **audio and symbolic** representations of music to **improve generation quality**.

A: Similar to my work - MusicConGen (ISIMIR'2024)

What are the qualities we want to improve with symbolic data?

- Controllability

What is the Symbolic Description?

Human understand music with notations and the conceptualized **informations**:

- BPM
- Meter
- Lead Sheet
 - Key
 - Chord
 - Melody
- Arrangement
- Structure
- MIDI
- Sheet Music
 - Staff and Tablature
- Genre
- Description (Autotagging)

Metrics for Evaluation

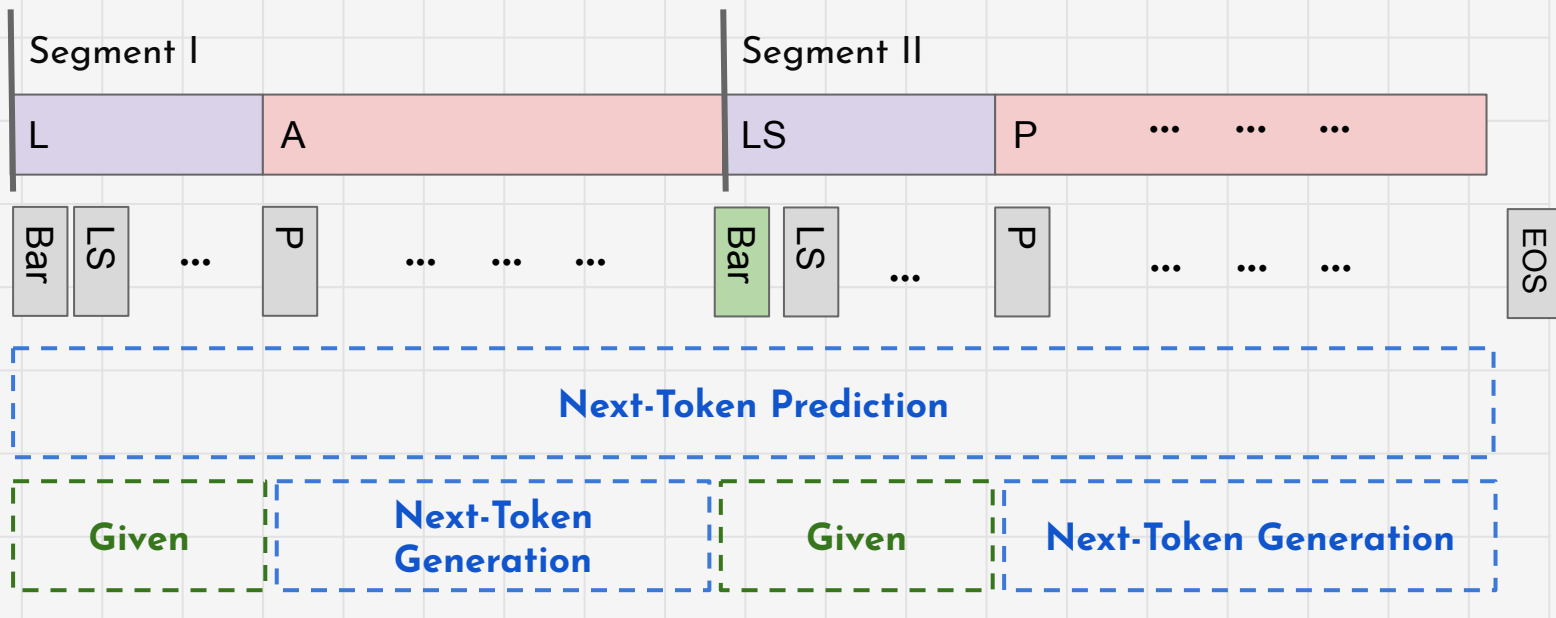
MIR eval Toolkit

- F-measure for BPM and Meter
- Chords
- FAD for audio quality

Combination of Different Dataset

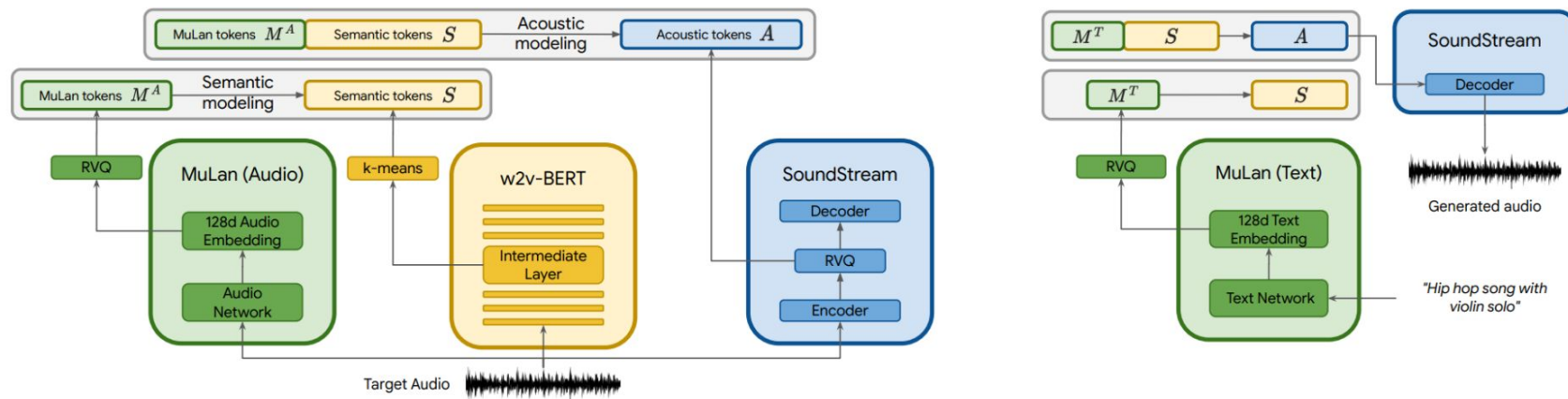
Method I:

- **Conditional Generation**, with decoder only (GPT-like) transformer
- **T5 Prefix-LM Mechanism**
 - Condition: Lead Sheet (L)
 - Generation: Acoustic Token (A) - can be generalized to multi-track



Method II: Hierarchical LLM

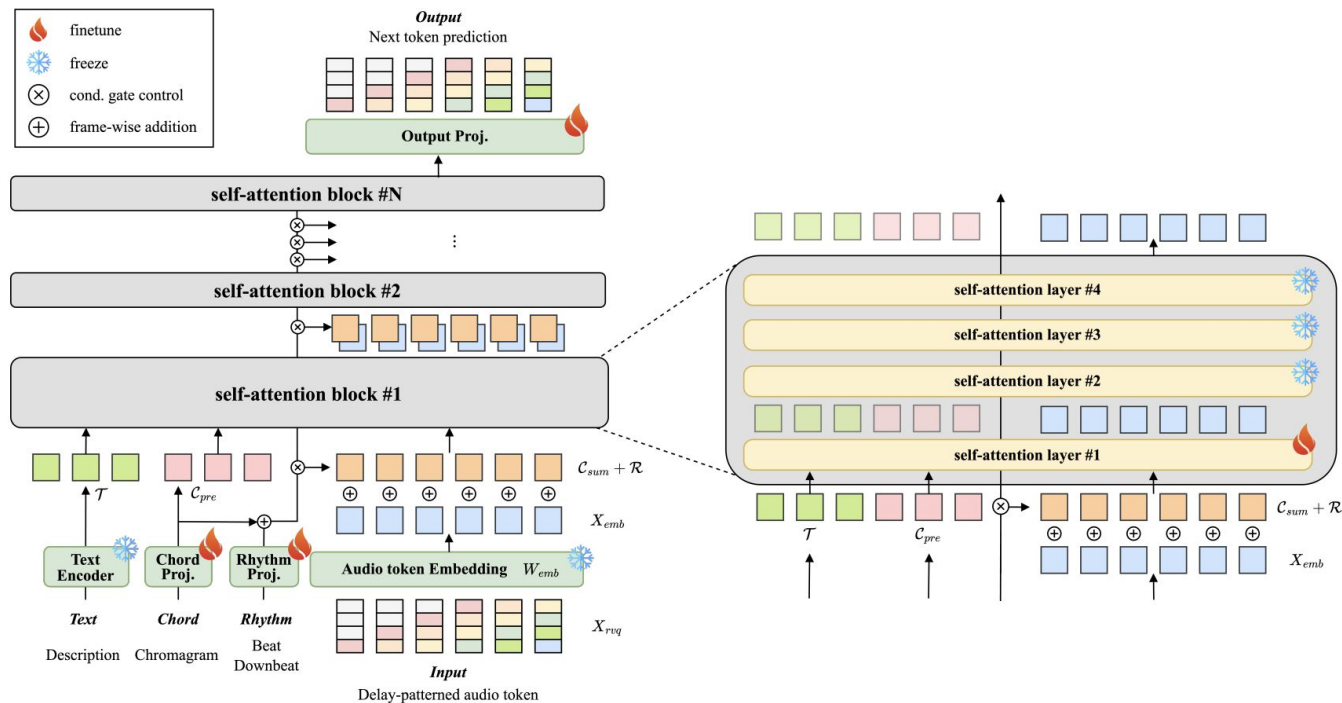
MusicLM: Generating Music From Text



Lead sheet -> Multi-track MIDI

Method III:

- Replacement of Positional/Sentence Embedding

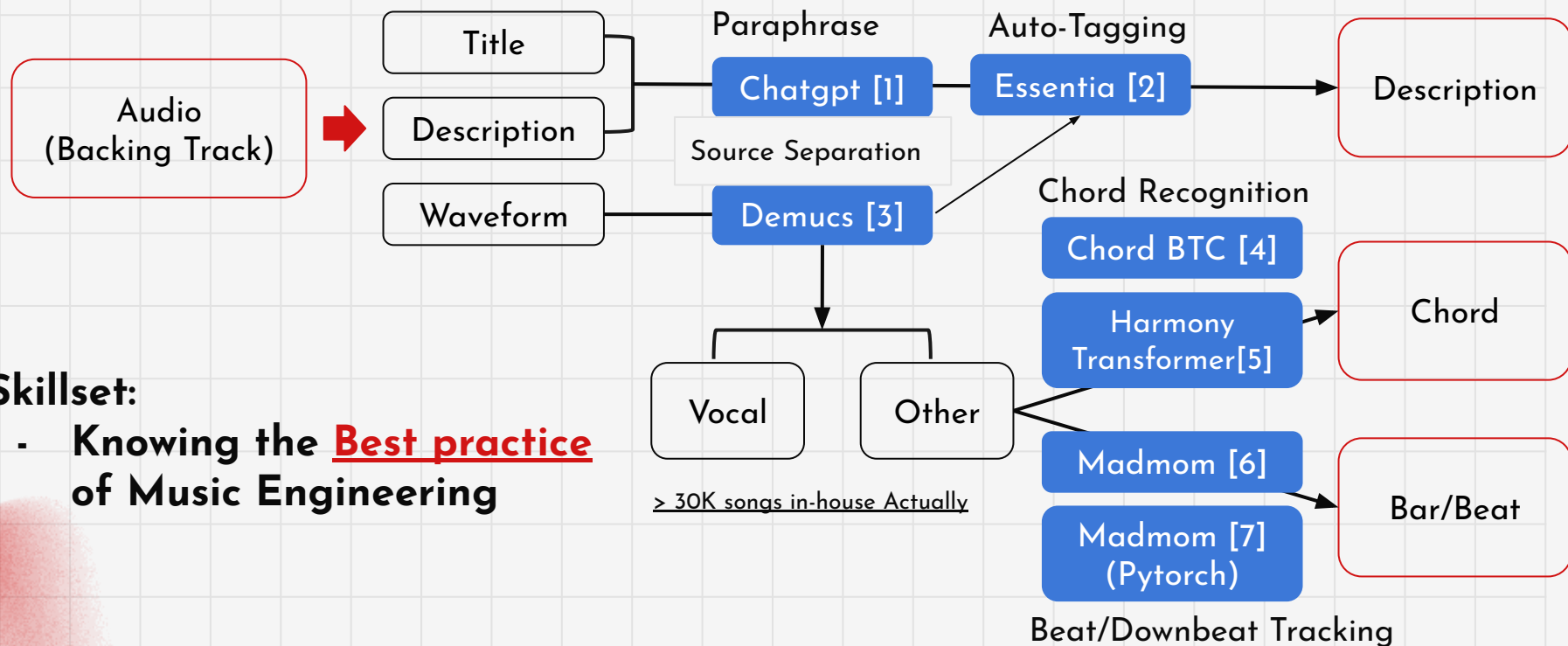


(a) MusiConGen model structure

(b) self-attention block

Dataset Building

- Pipeline from my work - MusicConGen (ISMIR'24)



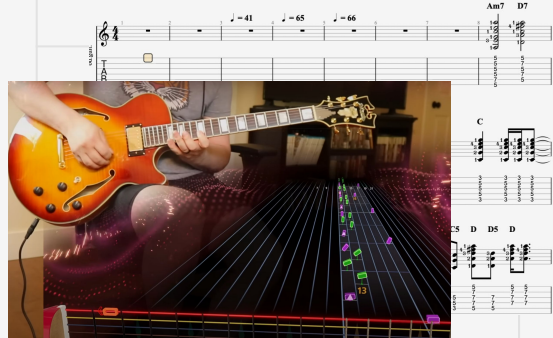
Skillset:

- Knowing the Best practice of Music Engineering

Dataset Building

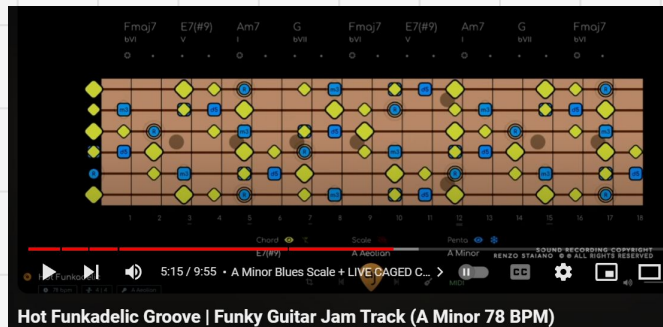
Highlights of My Inhouse Collection:

1. Data from Guitar Gaming Community



- Aligned audio and tab
- Finger position
- Chord label
- Over 1K songs
- Multi-track guitar
- Tab Generation
- Transcription

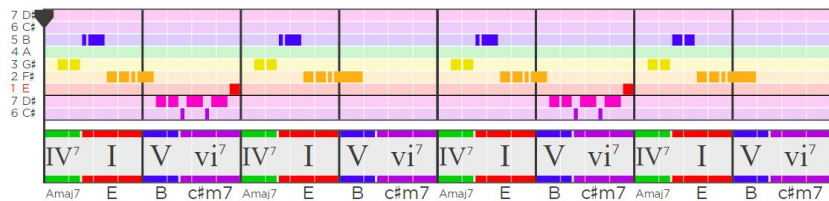
3. Backing Tracks



Skillset:

- Web Crawling, Data Cleaning
- Musicology

2. Lead Sheet from theorytab (108 stars)



- Over 30k songs
- Our backbone dataset of text2music model
- Description
- Key
- BPM
- Chord Progression
- High Quality after Curation (TODO)
- > Excellent Resources for any task!